

■ WORKED EXAMPLES ■

3.6 A Find bases and dimensions for all four fundamental subspaces if you know that

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 & 5 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} = LU = E^{-1}R.$$

By changing only *one* number in R , change the dimensions of all four subspaces.

Solution This matrix has pivots in columns 1 and 3. Its rank is $r = 2$.

Row space	Basis (1, 3, 0, 5) and (0, 0, 1, 6) from R . Dimension 2.
Column space	Basis (1, 2, 5) and (0, 1, 0) from E^{-1} (and A). Dimension 2.
Nullspace	Basis $(-3, 1, 0, 0)$ and $(-5, 0, -6, 1)$ from R . Dimension 2.
Nullspace of A^T	Basis $(-5, 0, 1)$ from row 3 of E . Dimension $3 - 2 = 1$.

We need to comment on that left nullspace $N(A^T)$. $EA = R$ says that the last row of E combines the three rows of A into the zero row of R . So that last row of E is a basis vector for the left nullspace. If R had *two* zero rows, then the last *two* rows of E would be a basis. (Just like elimination, $y^T A = 0^T$ combines rows of A to give zero rows in R .)

To change all these dimensions we need to change the rank r . One way to do that is to change an entry (*any entry*) in the zero row of R .

3.6 B Put four 1's into a 5 by 6 matrix of zeros, keeping the dimension of its *row space* as small as possible. Describe all the ways to make the dimension of its *column space* as small as possible. Describe all the ways to make the dimension of its *nullspace* as small as possible. How to make the *sum of the dimensions of all four subspaces small*?

Solution The rank is 1 if the four 1's go into the same row, or into the same column. They can also go into *two rows and two columns* (so $a_{ii} = a_{ij} = a_{ji} = a_{jj} = 1$). Since the column space and row space always have the same dimensions, this answers the first two questions: Dimension 1.

The nullspace has its smallest possible dimension $6 - 4 = 2$ when the rank is $r = 4$. To achieve rank 4, the 1's must go into four different rows and columns.

You can't do anything about the sum $r + (n - r) + r + (m - r) = n + m$. It will be $6 + 5 = 11$ no matter how the 1's are placed. The sum is 11 even if there aren't any 1's...

If all the other entries of A are 2's instead of 0's, how do these answers change?

Problem Set 3.6

- 1 (a) If a 7 by 9 matrix has rank 5, what are the dimensions of the four subspaces? What is the sum of all four dimensions?

(b) If a 3 by 4 matrix has rank 3, what are its column space and left nullspace?

2 Find bases and dimensions for the four subspaces associated with A and B :

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 8 \end{bmatrix}.$$

3 Find a basis for each of the four subspaces associated with A :

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

4 Construct a matrix with the required property or explain why this is impossible:

(a) Column space contains $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, row space contains $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

(b) Column space has basis $\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$, nullspace has basis $\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$.

(c) Dimension of nullspace = 1 + dimension of left nullspace.

(d) Left nullspace contains $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$, row space contains $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

(e) Row space = column space, nullspace $\neq$ left nullspace.

5 If V is the subspace spanned by $(1, 1, 1)$ and $(2, 1, 0)$, find a matrix A that has V as its row space. Find a matrix B that has V as its nullspace.

6 Without elimination, find dimensions and bases for the four subspaces for

$$A = \begin{bmatrix} 0 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}.$$

7 Suppose the 3 by 3 matrix A is invertible. Write down bases for the four subspaces for A , and also for the 3 by 6 matrix $B = [A \ A]$.

8 What are the dimensions of the four subspaces for A , B , and C , if I is the 3 by 3 identity matrix and 0 is the 3 by 2 zero matrix?

$$A = [I \ 0] \quad \text{and} \quad B = \begin{bmatrix} I & I \\ 0^T & 0^T \end{bmatrix} \quad \text{and} \quad C = [0].$$

9 Which subspaces are the same for these matrices of different sizes?

$$(a) [A] \quad \text{and} \quad \begin{bmatrix} A \\ A \end{bmatrix} \quad (b) \quad \begin{bmatrix} A \\ A \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} A & A \\ A & A \end{bmatrix}.$$

Prove that all three of those matrices have the *same rank* r .

- 10 If the entries of a 3 by 3 matrix are chosen randomly between 0 and 1, what are the most likely dimensions of the four subspaces? What if the matrix is 3 by 5?
- 11 (Important) A is an m by n matrix of rank r . Suppose there are right sides $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ has *no solution*.
- (a) What are all inequalities ($<$ or $\leq$) that must be true between m , n , and r ?
- (b) How do you know that $A^T\mathbf{y} = \mathbf{0}$ has solutions other than $\mathbf{y} = \mathbf{0}$?
- 12 Construct a matrix with $(1, 0, 1)$ and $(1, 2, 0)$ as a basis for its row space and its column space. Why can't this be a basis for the row space and nullspace?
- 13 True or false (with a reason or a counterexample):
- (a) If $m = n$ then the row space of A equals the column space.
- (b) The matrices A and $-A$ share the same four subspaces.
- (c) If A and B share the same four subspaces then A is a multiple of B .
- 14 Without computing A , find bases for its four fundamental subspaces:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 9 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

- 15 If you exchange the first two rows of A , which of the four subspaces stay the same? If $\mathbf{v} = (1, 2, 3, 4)$ is in the left nullspace of A , write down a vector in the left nullspace of the new matrix.
- 16 Explain why $\mathbf{v} = (1, 0, -1)$ *cannot be a row of A and also in the nullspace*.
- 17 Describe the four subspaces of $\mathbf{R}^3$ associated with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad I + A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

- 18 (Left nullspace) Add the extra column $\mathbf{b}$ and reduce A to echelon form:

$$[A \ \mathbf{b}] = \begin{bmatrix} 1 & 2 & 3 & b_1 \\ 4 & 5 & 6 & b_2 \\ 7 & 8 & 9 & b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & b_1 \\ 0 & -3 & -6 & b_2 - 4b_1 \\ 0 & 0 & 0 & b_3 - 2b_2 + b_1 \end{bmatrix}.$$

A combination of the rows of A has produced the zero row. What combination is it? (Look at $b_3 - 2b_2 + b_1$ on the right side.) Which vectors are in the nullspace of A^T and which are in the nullspace of A ?

- 19** Following the method of Problem 18, reduce A to echelon form and look at zero rows. The $\mathbf{b}$ column tells which combinations you have taken of the rows:

$$(a) \begin{bmatrix} 1 & 2 & b_1 \\ 3 & 4 & b_2 \\ 4 & 6 & b_3 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 2 & b_1 \\ 2 & 3 & b_2 \\ 2 & 4 & b_3 \\ 2 & 5 & b_4 \end{bmatrix}$$

From the $\mathbf{b}$ column after elimination, read off $m-r$ basis vectors in the left nullspace. Those $\mathbf{y}$'s are combinations of rows that give zero rows.

- 20** (a) Check that the solutions to $A\mathbf{x} = \mathbf{0}$ are perpendicular to the rows:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} = ER.$$

(b) How many independent solutions to $A^T \mathbf{y} = \mathbf{0}$? Why is $\mathbf{y}^T$ the last row of E^{-1} ?

- 21** Suppose A is the sum of two matrices of rank one: $A = \mathbf{u}\mathbf{v}^T + \mathbf{w}\mathbf{z}^T$.

- (a) Which vectors span the column space of A ?
 (b) Which vectors span the row space of A ?
 (c) The rank is less than 2 if _____ or if _____.
 (d) Compute A and its rank if $\mathbf{u} = \mathbf{z} = (1, 0, 0)$ and $\mathbf{v} = \mathbf{w} = (0, 0, 1)$.

- 22** Construct $A = \mathbf{u}\mathbf{v}^T + \mathbf{w}\mathbf{z}^T$ whose column space has basis $(1, 2, 4), (2, 2, 1)$ and whose row space has basis $(1, 0), (1, 1)$. Write A as (3 by 2) times (2 by 2).

- 23** Without multiplying matrices, find bases for the row and column spaces of A :

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} 3 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix}.$$

How do you know from these shapes that A cannot be invertible?

- 24** (Important) $A^T \mathbf{y} = \mathbf{d}$ is solvable when $\mathbf{d}$ is in which of the four subspaces? The solution $\mathbf{y}$ is unique when the _____ contains only the zero vector.
- 25** True or false (with a reason or a counterexample):
- (a) A and A^T have the same number of pivots.
 (b) A and A^T have the same left nullspace.
 (c) If the row space equals the column space then $A^T = A$.
 (d) If $A^T = -A$ then the row space of A equals the column space.

- 26 (**Rank of AB**) If $AB = C$, the rows of C are combinations of the rows of _____. So the rank of C is not greater than the rank of _____. Since $B^T A^T = C^T$, the rank of C is also not greater than the rank of _____.
- 27 If a, b, c are given with $a \neq 0$, how would you choose d so that $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has rank 1? Find a basis for the row space and nullspace. Show they are perpendicular!
- 28 Find the ranks of the 8 by 8 checkerboard matrix B and the chess matrix C :

$$B = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} r & n & b & q & k & b & n & r \\ p & p & p & p & p & p & p & p \\ \text{four zero rows} \\ p & p & p & p & p & p & p & p \\ r & n & b & q & k & b & n & r \end{bmatrix}$$

The numbers r, n, b, q, k, p are all different. Find bases for the row space and left nullspace of B and C . Challenge problem: Find a basis for the nullspace of C .

- 29 Can tic-tac-toe be completed (5 ones and 4 zeros in A) so that $\text{rank}(A) = 2$ but neither side passed up a winning move?

Challenge Problems

- 30 If $A = uv^T$ is a 2 by 2 matrix of rank 1, redraw Figure 3.5 to show clearly the Four Fundamental Subspaces. If B produces those same four subspaces, what is the exact relation of B to A ?
- 31 $\mathbf{M}$ is the space of 3 by 3 matrices. Multiply every matrix X in $\mathbf{M}$ by

$$A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}. \quad \text{Notice: } A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

(a) Which matrices X lead to $AX = \text{zero matrix}$?

(b) Which matrices have the form AX for some matrix X ?

(a) finds the “nullspace” of that operation AX and (b) finds the “column space”. What are the dimensions of those two subspaces of $\mathbf{M}$? Why do the dimensions add to $(n - r) + r = 9$?

- 32 Suppose the m by n matrices A and B have the same four subspaces. If they are both in row reduced echelon form, prove that F must equal G :

$$A = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} I & G \\ 0 & 0 \end{bmatrix}.$$

Chapter 4

Orthogonality

4.1 Orthogonality of the Four Subspaces

Two vectors are orthogonal when their dot product is zero: $v \cdot w = 0$ or $v^T w = 0$. This chapter moves to **orthogonal subspaces** and **orthogonal bases** and **orthogonal matrices**. The vectors in two subspaces, and the vectors in a basis, and the vectors in the columns, all pairs will be orthogonal. Think of $a^2 + b^2 = c^2$ for a *right triangle* with sides v and w .

Orthogonal vectors $v^T w = 0$ and $\|v\|^2 + \|w\|^2 = \|v + w\|^2$.

The right side is $(v + w)^T(v + w)$. This equals $v^T v + w^T w$ when $v^T w = w^T v = 0$.

Subspaces entered Chapter 3 to throw light on $Ax = b$. Right away we needed the column space (for b) and the nullspace (for x). Then the light turned onto A^T , uncovering two more subspaces. Those four fundamental subspaces reveal what a matrix really does.

A matrix multiplies a vector: A times x . At the first level this is only numbers. At the second level Ax is a combination of column vectors. The third level shows subspaces. But I don't think you have seen the whole picture until you study Figure 4.2. It fits the subspaces together, to show the hidden reality of A times x . The 90° angles between subspaces are new—and we have to say what those right angles mean.

The row space is perpendicular to the nullspace. Every row of A is perpendicular to every solution of $Ax = 0$. That gives the 90° angle on the left side of the figure. This perpendicularity of subspaces is Part 2 of the Fundamental Theorem of Linear Algebra.

The column space is perpendicular to the nullspace of A^T . When b is outside the column space—when we want to solve $Ax = b$ and can't do it—then this nullspace of A^T comes into its own. It contains the error $e = b - Ax$ in the “least-squares” solution. Least squares is the key application of linear algebra in this chapter.

Part 1 of the Fundamental Theorem gave the dimensions of the subspaces. The row and column spaces have the same dimension r (they are drawn the same size). The two nullspaces have the remaining dimensions $n - r$ and $m - r$. Now we will show that **the row space and nullspace are orthogonal subspaces inside $\mathbb{R}^n$** .

DEFINITION Two subspaces V and W of a vector space are *orthogonal* if every vector v in V is perpendicular to every vector w in W :

Orthogonal subspaces $v^T w = 0$ for all v in V and all w in W .

Example 1 The floor of your room (extended to infinity) is a subspace V . The line where two walls meet is a subspace W (one-dimensional). Those subspaces are orthogonal. Every vector up the meeting line is perpendicular to every vector in the floor.

Example 2 Two walls look perpendicular but they are not orthogonal subspaces! The meeting line is in both V and W —and this line is not perpendicular to itself. Two planes (dimensions 2 and 2 in $\mathbb{R}^3$) cannot be orthogonal subspaces.

When a vector is in two orthogonal subspaces, it *must* be zero. It is perpendicular to itself. It is v and it is w , so $v^T v = 0$. This has to be the zero vector.

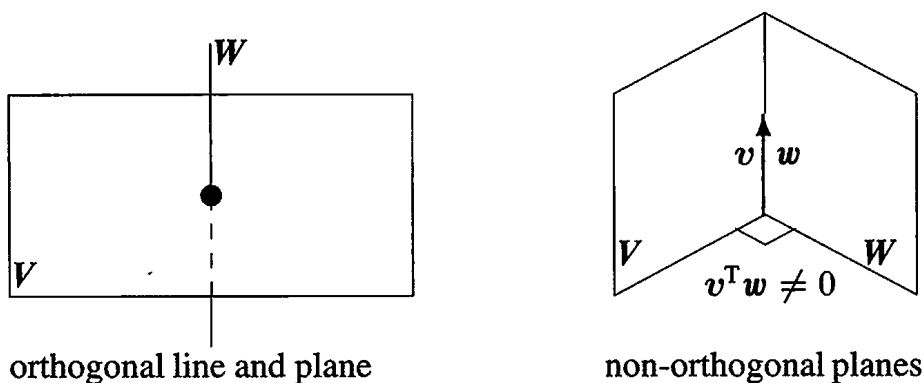


Figure 4.1: Orthogonality is impossible when $\dim V + \dim W > \text{dimension of whole space}$.

The crucial examples for linear algebra come from the fundamental subspaces. Zero is the only point where the nullspace meets the row space. More than that, the nullspace and row space of A meet at 90° . This key fact comes directly from $Ax = 0$:

Every vector x in the nullspace is perpendicular to every row of A , because $Ax = 0$. *The nullspace $N(A)$ and the row space $C(A^T)$ are orthogonal subspaces of $\mathbb{R}^n$.*

To see why x is perpendicular to the rows, look at $Ax = 0$. Each row multiplies x :

$$Ax = \begin{bmatrix} \text{row 1} \\ \vdots \\ \text{row } m \end{bmatrix} \begin{bmatrix} x \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \quad \begin{array}{l} \leftarrow (\text{row 1}) \cdot x \text{ is zero} \\ \leftarrow (\text{row } m) \cdot x \text{ is zero} \end{array} \quad (1)$$

The first equation says that row 1 is perpendicular to x . The last equation says that row m is perpendicular to x . *Every row has a zero dot product with x .* Then x is also perpendicular to every *combination* of the rows. The whole row space $C(A^T)$ is orthogonal to $N(A)$.

Here is a second proof of that orthogonality for readers who like matrix shorthand. The vectors in the row space are combinations $A^T y$ of the rows. Take the dot product of $A^T y$ with any x in the nullspace. *These vectors are perpendicular:*

$$\text{Nullspace and Row space} \quad x^T(A^T y) = (Ax)^T y = 0^T y = 0. \quad (2)$$

We like the first proof. You can see those rows of A multiplying x to produce zeros in equation (1). The second proof shows why A and A^T are both in the Fundamental Theorem. A^T goes with y and A goes with x . At the end we used $Ax = 0$.

Example 3 The rows of A are perpendicular to $x = (1, 1, -1)$ in the nullspace:

$$Ax = \begin{bmatrix} 1 & 3 & 4 \\ 5 & 2 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{gives the dot products} \quad \begin{array}{l} 1 + 3 - 4 = 0 \\ 5 + 2 - 7 = 0 \end{array}$$

Now we turn to the other two subspaces. In this example, the column space is all of $\mathbf{R}^2$. The nullspace of A^T is only the zero vector (orthogonal to every vector). The columns of A and nullspace of A^T are always orthogonal subspaces.

Every vector y in the nullspace of A^T is perpendicular to every column of A . *The left nullspace $N(A^T)$ and the column space $C(A)$ are orthogonal in $\mathbf{R}^m$.*

Apply the original proof to A^T . Its nullspace is orthogonal to its row space—and the row space of A^T is the column space of A . Q.E.D.

For a visual proof, look at $A^T y = 0$. Each column of A multiplies y to give 0:

$$C(A) \perp N(A^T) \quad A^T y = \begin{bmatrix} (\text{column } 1)^T \\ \vdots \\ (\text{column } n)^T \end{bmatrix} \begin{bmatrix} y \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}. \quad (3)$$

The dot product of y with every column of A is zero. Then y in the left nullspace is perpendicular to each column—and to the whole column space.

Orthogonal Complements

Important The fundamental subspaces are more than just orthogonal (in pairs). Their dimensions are also right. Two lines could be perpendicular in $\mathbf{R}^3$, but those lines *could not be* the row space and nullspace of a 3 by 3 matrix. The lines have dimensions 1 and 1, adding to 2. The correct dimensions r and $n - r$ must add to $n = 3$.

The fundamental subspaces have dimensions 2 and 1, or 3 and 0. Those subspaces are not only orthogonal, they are *orthogonal complements*.

DEFINITION The *orthogonal complement* of a subspace V contains *every* vector that is perpendicular to V . This orthogonal subspace is denoted by $V^\perp$ (pronounced “ V perp”).

By this definition, the nullspace is the orthogonal complement of the row space. *Every x that is perpendicular to the rows satisfies $Ax = 0$.*

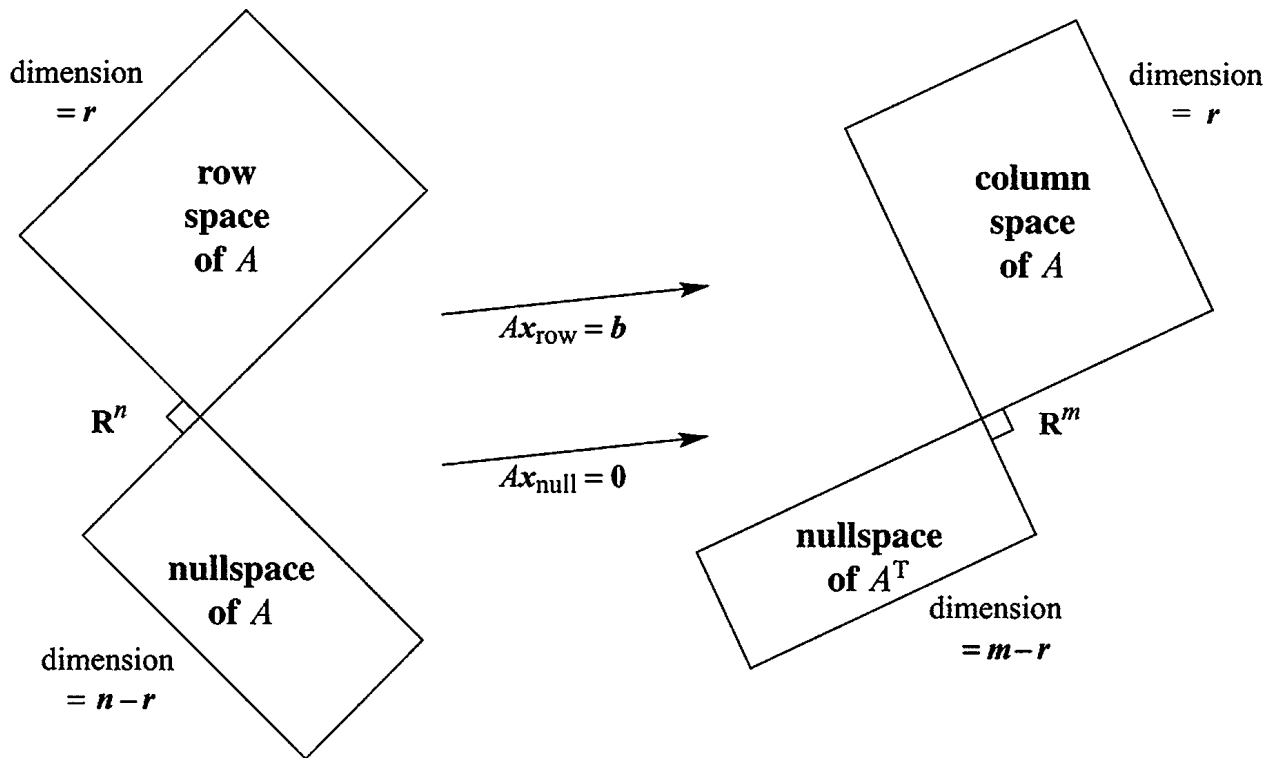


Figure 4.2: Two pairs of orthogonal subspaces. The dimensions add to n and add to m . **This is an important picture**—one pair of subspaces is in $\mathbb{R}^n$ and one pair is in $\mathbb{R}^m$.

The reverse is also true. *If v is orthogonal to the nullspace, it must be in the row space.* Otherwise we could add this v as an extra row of the matrix, without changing its nullspace. The row space would grow, which breaks the law $r + (n-r) = n$. We conclude that the nullspace complement $N(A)^\perp$ is exactly the row space $C(A^T)$.

The left nullspace and column space are orthogonal in $\mathbb{R}^m$, and they are orthogonal complements. Their dimensions r and $m-r$ add to the full dimension m .

Fundamental Theorem of Linear Algebra, Part 2

$N(A)$ is the orthogonal complement of the row space $C(A^T)$ (in $\mathbb{R}^n$).

$N(A^T)$ is the orthogonal complement of the column space $C(A)$ (in $\mathbb{R}^m$).

Part 1 gave the dimensions of the subspaces. Part 2 gives the 90° angles between them. The point of “complements” is that every x can be split into a *row space component* x_r and a *nullspace component* x_n . When A multiplies $x = x_r + x_n$, Figure 4.3 shows what happens:

The nullspace component goes to zero: $Ax_n = 0$.

The row space component goes to the column space: $Ax_r = Ax$.

Every vector goes to the column space! Multiplying by A cannot do anything else.

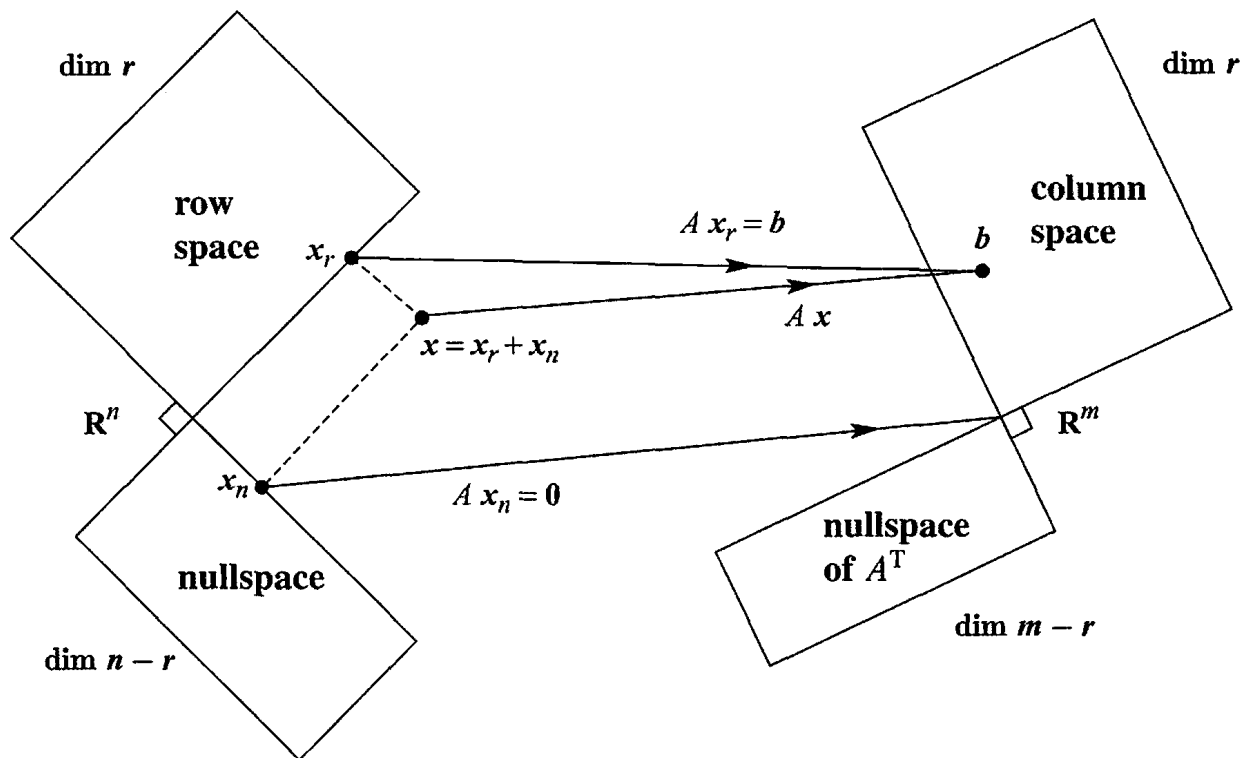


Figure 4.3: This update of Figure 4.2 shows the true action of A on $x = x_r + x_n$. Row space vector x_r to column space, nullspace vector x_n to zero.

More than that: *Every vector b in the column space comes from one and only one vector in the row space.* Proof: If $Ax_r = Ax'_r$, the difference $x_r - x'_r$ is in the nullspace. It is also in the row space, where x_r and x'_r came from. This difference must be the zero vector, because the nullspace and row space are perpendicular. Therefore $x_r = x'_r$.

There is an r by r invertible matrix hiding inside A , if we throw away the two nullspaces. **From the row space to the column space, A is invertible.** The “pseudoinverse” will invert it in Section 7.3.

Example 4 Every diagonal matrix has an r by r invertible submatrix:

$$A = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{contains the submatrix} \quad \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}.$$

The other eleven zeros are responsible for the nullspaces. The rank of B is also $r = 2$:

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 4 & 5 & 6 \\ 1 & 2 & 4 & 5 & 6 \end{bmatrix} \quad \text{contains} \quad \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} \quad \text{in the pivot rows and columns.}$$

Every A becomes a diagonal matrix, when we choose the right bases for $\mathbf{R}^n$ and $\mathbf{R}^m$. This **Singular Value Decomposition** has become extremely important in applications.

Combining Bases from Subspaces

What follows are some valuable facts about bases. They were saved until now—when we are ready to use them. After a week you have a clearer sense of what a basis is (*linearly independent* vectors that *span* the space). Normally we have to check both of these properties. When the count is right, one property implies the other:

Any n independent vectors in $\mathbf{R}^n$ must span $\mathbf{R}^n$. So they are a basis.

Any n vectors that span $\mathbf{R}^n$ must be independent. So they are a basis.

Starting with the correct number of vectors, one property of a basis produces the other. This is true in any vector space, but we care most about $\mathbf{R}^n$. When the vectors go into the columns of an n by n square matrix A , here are the same two facts:

If the n columns of A are independent, they span $\mathbf{R}^n$. So $Ax = b$ is solvable.

If the n columns span $\mathbf{R}^n$, they are independent. So $Ax = b$ has only one solution.

Uniqueness implies existence and existence implies uniqueness. **Then A is invertible.** If there are no free variables, the solution x is unique. There must be n pivots. Then back substitution solves $Ax = b$ (the solution exists).

Starting in the opposite direction, suppose $Ax = b$ can be solved for every b (*existence of solutions*). Then elimination produced no zero rows. There are n pivots and no free variables. The nullspace contains only $x = 0$ (*uniqueness of solutions*).

With bases for the row space and the nullspace, we have $r + (n - r) = n$ vectors. This is the right number. Those n vectors are independent.² *Therefore they span $\mathbf{R}^n$.*

Each x is the sum $x_r + x_n$ of a row space vector x_r and a nullspace vector x_n .

The splitting in Figure 4.3 shows the key point of orthogonal complements—the dimensions add to n and all vectors are fully accounted for.

Example 5 For $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ split $x = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ into $x_r + x_n = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

The vector $(2, 4)$ is in the row space. The orthogonal vector $(2, -1)$ is in the nullspace. The next section will compute this splitting for any A and x , by a projection.

²If a combination of all n vectors gives $x_r + x_n = 0$, then $x_r = -x_n$ is in both subspaces. So $x_r = x_n = 0$. All coefficients of the row space basis and nullspace basis must be zero—which proves independence of the n vectors together.

■ REVIEW OF THE KEY IDEAS ■

1. Subspaces V and W are orthogonal if every v in V is orthogonal to every w in W .
2. V and W are “orthogonal complements” if W contains **all** vectors perpendicular to V (and vice versa). Inside $\mathbf{R}^n$, the dimensions of complements V and W add to n .
3. The nullspace $N(A)$ and the row space $C(A^T)$ are orthogonal complements, from $Ax = \mathbf{0}$. Similarly $N(A^T)$ and $C(A)$ are orthogonal complements.
4. Any n independent vectors in $\mathbf{R}^n$ will span $\mathbf{R}^n$.
5. Every x in $\mathbf{R}^n$ has a nullspace component x_n and a row space component x_r .

■ WORKED EXAMPLES ■

4.1 A Suppose S is a six-dimensional subspace of nine-dimensional space $\mathbf{R}^9$.

- (a) What are the possible dimensions of subspaces orthogonal to S ?
- (b) What are the possible dimensions of the orthogonal complement $S^\perp$ of S ?
- (c) What is the smallest possible size of a matrix A that has row space S ?
- (d) What is the shape of its nullspace matrix N ?

Solution

- (a) If S is six-dimensional in $\mathbf{R}^9$, subspaces orthogonal to S can have dimensions 0, 1, 2, 3.
- (b) The complement $S^\perp$ is the largest orthogonal subspace, with dimension 3.
- (c) The smallest matrix A is 6 by 9 (its six rows are a basis for S).
- (d) Its nullspace matrix N is 9 by 3. The columns of N contain a basis for $S^\perp$.

If a new row 7 of B is a combination of the six rows of A , then B has the same row space as A . It also has the same nullspace matrix N . The special solutions s_1, s_2, s_3 will be the same. Elimination will change row 7 of B to all zeros.

4.1 B The equation $x - 3y - 4z = 0$ describes a plane P in $\mathbf{R}^3$ (actually a subspace).

- (a) The plane P is the nullspace $N(A)$ of what 1 by 3 matrix A ?
- (b) Find a basis s_1, s_2 of special solutions of $x - 3y - 4z = 0$ (these would be the columns of the nullspace matrix N).

- (c) Also find a basis for the line $P^\perp$ that is perpendicular to P .
- (d) Split $v = (6, 4, 5)$ into its nullspace component v_n in P and its row space component v_r in $P^\perp$.

Solution

- (a) The equation $x - 3y - 4z = 0$ is $Ax = 0$ for the 1 by 3 matrix $A = [1 \ -3 \ -4]$.
- (b) Columns 2 and 3 are free (the only pivot is 1). The special solutions with free variables 1 and 0 are $s_1 = (3, 1, 0)$ and $s_2 = (4, 0, 1)$ in the plane $P = N(A)$.
- (c) The row space of A is the line $P^\perp$ in the direction of the row $z = (1, -3, -4)$.
- (d) To split v into $v_n + v_r = (c_1s_1 + c_2s_2) + c_3z$, solve for $c_1 = 1, c_2 = 1, c_3 = -1$.

$$\begin{bmatrix} 6 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 1 \\ 1 & 0 & -3 \\ 0 & 1 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \quad \begin{array}{l} v_n = s_1 + s_2 = (7, 1, 1) \text{ is in } P = N(A) \\ v_r = -s_3 = (-1, 3, 4) \text{ is in } P^\perp = C(A^T) \\ v = (6, 4, 5) \text{ equals } (7, 1, 1) + (-1, 3, 4) \end{array}$$

This method used a basis for each subspace combined into an overall basis s_1, s_2, z . Section 4.2 will also project v onto a subspace S . There we will not need a basis for the perpendicular subspace $S^\perp$.

Problem Set 4.1

Questions 1–12 grow out of Figures 4.2 and 4.3 with four subspaces.

- Construct any 2 by 3 matrix of rank one. Copy Figure 4.2 and put one vector in each subspace (two in the nullspace). Which vectors are orthogonal?
- Redraw Figure 4.3 for a 3 by 2 matrix of rank $r = 2$. Which subspace is Z (zero vector only)? The nullspace part of any vector x in $\mathbf{R}^2$ is $x_n = \underline{\hspace{2cm}}$.
- Construct a matrix with the required property or say why that is impossible:
 - Column space contains $\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$, nullspace contains $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$
 - Row space contains $\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$, nullspace contains $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$
 - $Ax = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ has a solution and $A^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 - Every row is orthogonal to every column (A is not the zero matrix)
 - Columns add up to a column of zeros, rows add to a row of 1's.
- If $AB = 0$ then the columns of B are in the $\underline{\hspace{2cm}}$ of A . The rows of A are in the $\underline{\hspace{2cm}}$ of B . Why can't A and B be 3 by 3 matrices of rank 2?

- 5 (a) If $Ax = b$ has a solution and $A^T y = 0$, is $(y^T x = 0)$ or $(y^T b = 0)$?
 (b) If $A^T y = (1, 1, 1)$ has a solution and $Ax = 0$, then ____.
- 6 This system of equations $Ax = b$ has *no solution* (they lead to $0 = 1$):

$$\begin{aligned} x + 2y + 2z &= 5 \\ 2x + 2y + 3z &= 5 \\ 3x + 4y + 5z &= 9 \end{aligned}$$

Find numbers y_1, y_2, y_3 to multiply the equations so they add to $0 = 1$. You have found a vector y in which subspace? Its dot product $y^T b$ is 1, so no solution x .

- 7 Every system with no solution is like the one in Problem 6. There are numbers $y_1, \dots, y_m$ that multiply the m equations so they add up to $0 = 1$. This is called **Fredholm's Alternative**:

Exactly one of these problems has a solution

$$Ax = b \quad \text{OR} \quad A^T y = 0 \quad \text{with} \quad y^T b = 1.$$

If b is not in the column space of A , it is not orthogonal to the nullspace of A^T . Multiply the equations $x_1 - x_2 = 1$ and $x_2 - x_3 = 1$ and $x_1 - x_3 = 1$ by numbers y_1, y_2, y_3 chosen so that the equations add up to $0 = 1$.

- 8 In Figure 4.3, how do we know that Ax_r is equal to Ax ? How do we know that this vector is in the column space? If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ what is x_r ?
- 9 If $A^T Ax = 0$ then $Ax = 0$. Reason: Ax is in the nullspace of A^T and also in the ____ of A and those spaces are _____. Conclusion: $A^T A$ has the same nullspace as A . This key fact is repeated in the next section.
- 10 Suppose A is a symmetric matrix ($A^T = A$).
- (a) Why is its column space perpendicular to its nullspace?
- (b) If $Ax = 0$ and $Az = 5z$, which subspaces contain these "eigenvectors" x and z ? **Symmetric matrices have perpendicular eigenvectors** $x^T z = 0$.
- 11 (Recommended) Draw Figure 4.2 to show each subspace correctly for

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}.$$

- 12 Find the pieces x_r and x_n and draw Figure 4.3 properly if

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad x = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Questions 13–23 are about orthogonal subspaces.

- 13 Put bases for the subspaces V and W into the columns of matrices V and W . Explain why the test for orthogonal subspaces can be written $V^T W = \text{zero matrix}$. This matches $v^T w = 0$ for orthogonal vectors.
- 14 The floor V and the wall W are not orthogonal subspaces, because they share a nonzero vector (along the line where they meet). No planes V and W in $\mathbf{R}^3$ can be orthogonal! Find a vector in the column spaces of both matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 5 & 4 \\ 6 & 3 \\ 5 & 1 \end{bmatrix}$$

This will be a vector Ax and also $B\hat{x}$. Think 3 by 4 with the matrix $\begin{bmatrix} A & B \end{bmatrix}$.

- 15 Extend Problem 14 to a p -dimensional subspace V and a q -dimensional subspace W of $\mathbf{R}^n$. What inequality on $p + q$ guarantees that V intersects W in a nonzero vector? These subspaces cannot be orthogonal.
- 16 Prove that every y in $N(A^T)$ is perpendicular to every Ax in the column space, using the matrix shorthand of equation (2). Start from $A^T y = \mathbf{0}$.
- 17 If S is the subspace of $\mathbf{R}^3$ containing only the zero vector, what is $S^\perp$? If S is spanned by $(1, 1, 1)$, what is $S^\perp$? If S is spanned by $(1, 1, 1)$ and $(1, 1, -1)$, what is a basis for $S^\perp$?
- 18 Suppose S only contains two vectors $(1, 5, 1)$ and $(2, 2, 2)$ (not a subspace). Then $S^\perp$ is the nullspace of the matrix $A = \underline{\hspace{2cm}}$. $S^\perp$ is a subspace even if S is not.
- 19 Suppose L is a one-dimensional subspace (a line) in $\mathbf{R}^3$. Its orthogonal complement $L^\perp$ is the perpendicular to L . Then $(L^\perp)^\perp$ is a perpendicular to $L^\perp$. In fact $(L^\perp)^\perp$ is the same as .
- 20 Suppose V is the whole space $\mathbf{R}^4$. Then $V^\perp$ contains only the vector . Then $(V^\perp)^\perp$ is . So $(V^\perp)^\perp$ is the same as .
- 21 Suppose S is spanned by the vectors $(1, 2, 2, 3)$ and $(1, 3, 3, 2)$. Find two vectors that span $S^\perp$. This is the same as solving $Ax = \mathbf{0}$ for which A ?
- 22 If P is the plane of vectors in $\mathbf{R}^4$ satisfying $x_1 + x_2 + x_3 + x_4 = 0$, write a basis for $P^\perp$. Construct a matrix that has P as its nullspace.
- 23 If a subspace S is contained in a subspace V , prove that $S^\perp$ contains $V^\perp$.

Questions 24–30 are about perpendicular columns and rows.

- 24** Suppose an n by n matrix is invertible: $AA^{-1} = I$. Then the first column of A^{-1} is orthogonal to the space spanned by which rows of A ?
- 25** Find $A^T A$ if the columns of A are unit vectors, all mutually perpendicular.
- 26** Construct a 3 by 3 matrix A with no zero entries whose columns are mutually perpendicular. Compute $A^T A$. Why is it a diagonal matrix?
- 27** The lines $3x + y = b_1$ and $6x + 2y = b_2$ are _____. They are the same line if _____. In that case (b_1, b_2) is perpendicular to the vector _____. The nullspace of the matrix is the line $3x + y =$ _____. One particular vector in that nullspace is _____.
- 28** Why is each of these statements false?
- (a) $(1, 1, 1)$ is perpendicular to $(1, 1, -2)$ so the planes $x + y + z = 0$ and $x + y - 2z = 0$ are orthogonal subspaces.
 - (b) The subspace spanned by $(1, 1, 0, 0, 0)$ and $(0, 0, 0, 1, 1)$ is the orthogonal complement of the subspace spanned by $(1, -1, 0, 0, 0)$ and $(2, -2, 3, 4, -4)$.
 - (c) Two subspaces that meet only in the zero vector are orthogonal.
- 29** Find a matrix with $v = (1, 2, 3)$ in the row space and column space. Find another matrix with v in the nullspace and column space. Which pairs of subspaces can v not be in?

Challenge Problems

- 30** Suppose A is 3 by 4 and B is 4 by 5 and $AB = 0$. So $N(A)$ contains $C(B)$. Prove from the dimensions of $N(A)$ and $C(B)$ that $\text{rank}(A) + \text{rank}(B) \leq 4$.
- 31** The command $N = \text{null}(A)$ will produce a basis for the nullspace of A . Then the command $B = \text{null}(N')$ will produce a basis for the _____ of A .
- 32** Suppose I give you four nonzero vectors r, n, c, l in $\mathbb{R}^2$.
- (a) What are the conditions for those to be bases for the four fundamental subspaces $C(A^T)$, $N(A)$, $C(A)$, $N(A^T)$ of a 2 by 2 matrix?
 - (b) What is one possible matrix A ?
- 33** Suppose I give you eight vectors $r_1, r_2, n_1, n_2, c_1, c_2, l_1, l_2$ in $\mathbb{R}^4$.
- (a) What are the conditions for those pairs to be bases for the four fundamental subspaces of a 4 by 4 matrix?
 - (b) What is one possible matrix A ?

4.2 Projections

May we start this section with two questions? (In addition to that one.) The first question aims to show that projections are easy to visualize. The second question is about “projection matrices”—symmetric matrices with $P^2 = P$. *The projection of $\mathbf{b}$ is $P\mathbf{b}$.*

- 1 What are the projections of $\mathbf{b} = (2, 3, 4)$ onto the z axis and the xy plane?
- 2 What matrices produce those projections onto a line and a plane?

When $\mathbf{b}$ is projected onto a line, *its projection $\mathbf{p}$ is the part of $\mathbf{b}$ along that line.* If $\mathbf{b}$ is projected onto a plane, $\mathbf{p}$ is the part in that plane. *The projection $\mathbf{p}$ is $P\mathbf{b}$.*

The projection matrix P multiplies $\mathbf{b}$ to give $\mathbf{p}$. This section finds $\mathbf{p}$ and P .

The projection onto the z axis we call $\mathbf{p}_1$. The second projection drops straight down to the xy plane. The picture in your mind should be Figure 4.4. Start with $\mathbf{b} = (2, 3, 4)$. One projection gives $\mathbf{p}_1 = (0, 0, 4)$ and the other gives $\mathbf{p}_2 = (2, 3, 0)$. Those are the parts of $\mathbf{b}$ along the z axis and in the xy plane.

The projection matrices P_1 and P_2 are 3 by 3. They multiply $\mathbf{b}$ with 3 components to produce $\mathbf{p}$ with 3 components. Projection onto a line comes from a rank one matrix. Projection onto a plane comes from a rank two matrix:

$$\text{Onto the } z \text{ axis: } P_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{Onto the } xy \text{ plane: } P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

P_1 picks out the z component of every vector. P_2 picks out the x and y components. To find the projections $\mathbf{p}_1$ and $\mathbf{p}_2$ of $\mathbf{b}$, multiply $\mathbf{b}$ by P_1 and P_2 (small $\mathbf{p}$ for the vector, capital P for the matrix that produces it):

$$\mathbf{p}_1 = P_1 \mathbf{b} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} \quad \mathbf{p}_2 = P_2 \mathbf{b} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}.$$

In this case the projections $\mathbf{p}_1$ and $\mathbf{p}_2$ are perpendicular. The xy plane and the z axis are *orthogonal subspaces*, like the floor of a room and the line between two walls.

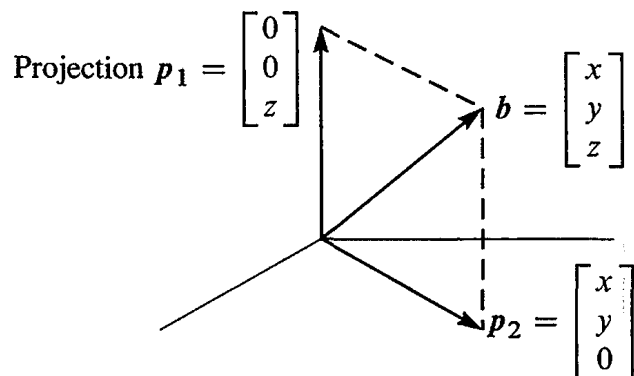


Figure 4.4: The projections $\mathbf{p}_1 = P_1 \mathbf{b}$ and $\mathbf{p}_2 = P_2 \mathbf{b}$ onto the z axis and the xy plane.

More than that, the line and plane are orthogonal **complements**. Their dimensions add to $1 + 2 = 3$. Every vector $\mathbf{b}$ in the whole space is the sum of its parts in the two subspaces. The projections $\mathbf{p}_1$ and $\mathbf{p}_2$ are exactly those parts:

$$\text{The vectors give } \mathbf{p}_1 + \mathbf{p}_2 = \mathbf{b}. \quad \text{The matrices give } P_1 + P_2 = I. \quad (1)$$

This is perfect. Our goal is reached—for this example. We have the same goal for any line and any plane and any n -dimensional subspace. The object is to find the part $\mathbf{p}$ in each subspace, and the projection matrix P that produces that part $\mathbf{p} = P\mathbf{b}$. Every subspace of $\mathbf{R}^m$ has its own m by m projection matrix. To compute P , we absolutely need a good description of the subspace that it projects onto.

The best description of a subspace is a basis. We put the basis vectors into the columns of A . *Now we are projecting onto the column space of A !* Certainly the z axis is the column space of the 3 by 1 matrix A_1 . The xy plane is the column space of A_2 . That plane is also the column space of A_3 (a subspace has many bases):

$$A_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad A_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad A_3 = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 0 & 0 \end{bmatrix}.$$

Our problem is *to project any $\mathbf{b}$ onto the column space of any m by n matrix*. Start with a line (dimension $n = 1$). The matrix A has only one column. Call it $\mathbf{a}$.

Projection Onto a Line

A line goes through the origin in the direction of $\mathbf{a} = (a_1, \dots, a_m)$. Along that line, we want the point $\mathbf{p}$ closest to $\mathbf{b} = (b_1, \dots, b_m)$. The key to projection is orthogonality: *The line from $\mathbf{b}$ to $\mathbf{p}$ is perpendicular to the vector $\mathbf{a}$.* This is the dotted line marked e for error in Figure 4.5—which we now compute by algebra.

The projection $\mathbf{p}$ is some multiple of $\mathbf{a}$. Call it $\mathbf{p} = \hat{x}\mathbf{a} = \text{"x hat"} \text{ times } \mathbf{a}$. Computing this number $\hat{x}$ will give the vector $\mathbf{p}$. Then from the formula for $\mathbf{p}$, we read off the projection matrix P . These three steps will lead to all projection matrices: *find $\hat{x}$, then find the vector $\mathbf{p}$, then find the matrix P .*

The dotted line $\mathbf{b} - \mathbf{p}$ is $\mathbf{e} = \mathbf{b} - \hat{x}\mathbf{a}$. It is perpendicular to $\mathbf{a}$ —this will determine $\hat{x}$. Use the fact that $\mathbf{b} - \mathbf{p}$ is perpendicular to $\mathbf{a}$ when their dot product is zero:

$$\begin{array}{l} \text{Projecting } \mathbf{b} \text{ onto } \mathbf{a}, \text{ error } \mathbf{e} = \mathbf{b} - \hat{x}\mathbf{a} \\ \mathbf{a} \cdot (\mathbf{b} - \hat{x}\mathbf{a}) = 0 \quad \text{or} \quad \mathbf{a} \cdot \mathbf{b} - \hat{x}\mathbf{a} \cdot \mathbf{a} = 0 \end{array} \quad \hat{x} = \frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}} = \frac{\mathbf{a}^T \mathbf{b}}{\mathbf{a}^T \mathbf{a}}. \quad (2)$$

The multiplication $\mathbf{a}^T \mathbf{b}$ is the same as $\mathbf{a} \cdot \mathbf{b}$. Using the transpose is better, because it applies also to matrices. Our formula $\hat{x} = \mathbf{a}^T \mathbf{b} / \mathbf{a}^T \mathbf{a}$ gives the projection $\mathbf{p} = \hat{x}\mathbf{a}$.

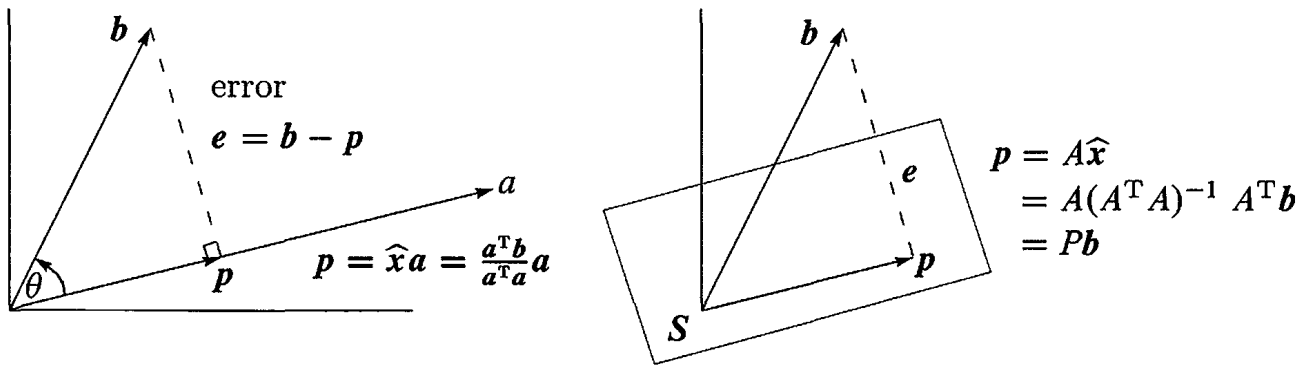


Figure 4.5: The projection p of b onto a line and onto $S = \text{column space of } A$.

The projection of b onto the line through a is the vector $p = \hat{x}a = \frac{a^T b}{a^T a}a$.

Special case 1: If $b = a$ then $\hat{x} = 1$. The projection of a onto a is itself. $Pa = a$.

Special case 2: If b is perpendicular to a then $a^T b = 0$. The projection is $p = 0$.

Example 1 Project $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ onto $a = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ to find $p = \hat{x}a$ in Figure 4.5.

Solution The number $\hat{x}$ is the ratio of $a^T b = 5$ to $a^T a = 9$. So the projection is $p = \frac{5}{9}a$. The error vector between b and p is $e = b - p$. Those vectors p and e will add to $b = (1, 1, 1)$:

$$p = \frac{5}{9}a = \left(\frac{5}{9}, \frac{10}{9}, \frac{10}{9}\right) \quad \text{and} \quad e = b - p = \left(\frac{4}{9}, -\frac{1}{9}, -\frac{1}{9}\right).$$

The error e should be perpendicular to $a = (1, 2, 2)$ and it is: $e^T a = \frac{4}{9} - \frac{2}{9} - \frac{2}{9} = 0$.

Look at the right triangle of b , p , and e . The vector b is split into two parts—its component along the line is p , its perpendicular part is e . Those two sides of a right triangle have length $\|b\| \cos \theta$ and $\|b\| \sin \theta$. Trigonometry matches the dot product:

$$p = \frac{a^T b}{a^T a}a \quad \text{has length} \quad \|p\| = \frac{\|a\| \|b\| \cos \theta}{\|a\|^2} \|a\| = \|b\| \cos \theta. \quad (3)$$

The dot product is a lot simpler than getting involved with $\cos \theta$ and the length of b . The example has square roots in $\cos \theta = 5/3\sqrt{3}$ and $\|b\| = \sqrt{3}$. There are no square roots in the projection $p = 5a/9$. The good way to $5/9$ is $b^T a / a^T a$.

Now comes the **projection matrix**. In the formula for p , what matrix is multiplying b ? You can see the matrix better if the number $\hat{x}$ is on the right side of a :

Projection matrix P $p = a\hat{x} = a \frac{a^T b}{a^T a} = Pb \quad \text{when the matrix is} \quad P = \frac{aa^T}{a^T a}.$

P is a column times a row! The column is $\mathbf{a}$, the row is $\mathbf{a}^T$. Then divide by the number $\mathbf{a}^T \mathbf{a}$. The projection matrix P is m by m , but *its rank is one*. We are projecting onto a one-dimensional subspace, the line through $\mathbf{a}$. That is the column space of P .

Example 2 Find the projection matrix $P = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T \mathbf{a}}$ onto the line through $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$.

Solution Multiply column $\mathbf{a}$ times row $\mathbf{a}^T$ and divide by $\mathbf{a}^T \mathbf{a} = 9$:

$$\text{Projection matrix} \quad P = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T \mathbf{a}} = \frac{1}{9} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}.$$

This matrix projects *any* vector $\mathbf{b}$ onto $\mathbf{a}$. Check $\mathbf{p} = P\mathbf{b}$ for $\mathbf{b} = (1, 1, 1)$ in Example 1:

$$\mathbf{p} = P\mathbf{b} = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 5 \\ 10 \\ 10 \end{bmatrix} \quad \text{which is correct.}$$

If the vector $\mathbf{a}$ is doubled, the matrix P stays the same. It still projects onto the same line. If the matrix is squared, P^2 equals P . **Projecting a second time doesn't change anything**, so $P^2 = P$. The diagonal entries of P add up to $\frac{1}{9}(1 + 4 + 4) = 1$.

The matrix $I - P$ should be a projection too. It produces the other side $\mathbf{e}$ of the triangle—the perpendicular part of $\mathbf{b}$. Note that $(I - P)\mathbf{b}$ equals $\mathbf{b} - \mathbf{p}$ which is $\mathbf{e}$ in the left nullspace. **When P projects onto one subspace, $I - P$ projects onto the perpendicular subspace.** Here $I - P$ projects onto the plane perpendicular to $\mathbf{a}$.

Now we move beyond projection onto a line. Projecting onto an n -dimensional subspace of $\mathbf{R}^m$ takes more effort. The crucial formulas will be collected in equations (5)–(6)–(7). Basically you need to remember those three equations.

Projection Onto a Subspace

Start with n vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$ in $\mathbf{R}^m$. Assume that these $\mathbf{a}$'s are linearly independent.

Problem: Find the combination $\mathbf{p} = \hat{x}_1 \mathbf{a}_1 + \dots + \hat{x}_n \mathbf{a}_n$ closest to a given vector $\mathbf{b}$. We are projecting each $\mathbf{b}$ in $\mathbf{R}^m$ onto the subspace spanned by the $\mathbf{a}$'s, to get $\mathbf{p}$.

With $n = 1$ (only one vector $\mathbf{a}_1$) this is projection onto a line. The line is the column space of A , which has just one column. In general the matrix A has n columns $\mathbf{a}_1, \dots, \mathbf{a}_n$.

The combinations in $\mathbf{R}^m$ are the vectors $A\mathbf{x}$ in the column space. We are looking for the particular combination $\mathbf{p} = A\hat{\mathbf{x}}$ (**the projection**) that is closest to $\mathbf{b}$. The hat over $\hat{\mathbf{x}}$ indicates the *best* choice $\hat{\mathbf{x}}$, to give the closest vector in the column space. That choice is $\mathbf{a}^T \mathbf{b} / \mathbf{a}^T \mathbf{a}$ when $n = 1$. For $n > 1$, the best $\hat{\mathbf{x}}$ is to be found now.

We compute projections onto n -dimensional subspaces in three steps as before: **Find the vector $\hat{\mathbf{x}}$, find the projection $\mathbf{p} = A\hat{\mathbf{x}}$, find the matrix P .**

The key is in the geometry! The dotted line in Figure 4.5 goes from $\mathbf{b}$ to the nearest point $A\hat{\mathbf{x}}$ in the subspace. **This error vector $\mathbf{b} - A\hat{\mathbf{x}}$ is perpendicular to the subspace.**

The error $\mathbf{b} - A\hat{\mathbf{x}}$ makes a right angle with all the vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$. The n right angles give the n equations for $\hat{\mathbf{x}}$:

$$\begin{array}{l} \mathbf{a}_1^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \\ \vdots \\ \mathbf{a}_n^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \end{array} \quad \text{or} \quad \begin{bmatrix} -\mathbf{a}_1^T \\ \vdots \\ -\mathbf{a}_n^T \end{bmatrix} \begin{bmatrix} \mathbf{b} - A\hat{\mathbf{x}} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}. \quad (4)$$

The matrix with those rows $\mathbf{a}_i^T$ is A^T . The n equations are exactly $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$.

Rewrite $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$ in its famous form $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$. This is the equation for $\hat{\mathbf{x}}$, and the coefficient matrix is $A^T A$. Now we can find $\hat{\mathbf{x}}$ and $\mathbf{p}$ and P , in that order:

The combination $\mathbf{p} = \hat{x}_1 \mathbf{a}_1 + \dots + \hat{x}_n \mathbf{a}_n = A\hat{\mathbf{x}}$ that is closest to $\mathbf{b}$ comes from

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0} \quad \text{or} \quad A^T A \hat{\mathbf{x}} = A^T \mathbf{b}. \quad (5)$$

This symmetric matrix $A^T A$ is n by n . It is invertible if the $\mathbf{a}$'s are independent. The solution is $\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$. The *projection* of $\mathbf{b}$ onto the subspace is $\mathbf{p}$:

$$\mathbf{p} = A\hat{\mathbf{x}} = A(A^T A)^{-1} A^T \mathbf{b}. \quad (6)$$

This formula shows the n by n *projection matrix* that produces $\mathbf{p} = P\mathbf{b}$:

$$P = A(A^T A)^{-1} A^T. \quad (7)$$

Compare with projection onto a line, when the matrix A has only one column $\mathbf{a}$:

$$\text{For } n = 1 \quad \hat{x} = \frac{\mathbf{a}^T \mathbf{b}}{\mathbf{a}^T \mathbf{a}} \quad \text{and} \quad \mathbf{p} = \mathbf{a} \frac{\mathbf{a}^T \mathbf{b}}{\mathbf{a}^T \mathbf{a}} \quad \text{and} \quad P = \frac{\mathbf{a} \mathbf{a}^T}{\mathbf{a}^T \mathbf{a}}.$$

Those formulas are identical with (5) and (6) and (7). The number $\mathbf{a}^T \mathbf{a}$ becomes the matrix $A^T A$. When it is a number, we divide by it. When it is a matrix, we invert it. The new formulas contain $(A^T A)^{-1}$ instead of $1/\mathbf{a}^T \mathbf{a}$. The linear independence of the columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ will guarantee that this inverse matrix exists.

The key step was $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$. We used geometry ($\mathbf{e}$ is perpendicular to all the $\mathbf{a}$'s). Linear algebra gives this "normal equation" too, in a very quick way:

1. Our subspace is the column space of A .
2. The error vector $\mathbf{b} - A\hat{\mathbf{x}}$ is perpendicular to that column space.
3. Therefore $\mathbf{b} - A\hat{\mathbf{x}}$ is in the nullspace of A^T . This means $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$.

The left nullspace is important in projections. That nullspace of A^T contains the error vector $\mathbf{e} = \mathbf{b} - A\hat{\mathbf{x}}$. The vector $\mathbf{b}$ is being split into the projection $\mathbf{p}$ and the error $\mathbf{e} = \mathbf{b} - \mathbf{p}$. Projection produces a right triangle (Figure 4.5) with sides $\mathbf{p}$, $\mathbf{e}$, and $\mathbf{b}$.

Example 3 If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$ find $\hat{\mathbf{x}}$ and $\mathbf{p}$ and P .

Solution Compute the square matrix $A^T A$ and also the vector $A^T \mathbf{b}$:

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \quad \text{and} \quad A^T \mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}.$$

Now solve the normal equation $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ to find $\hat{\mathbf{x}}$:

$$\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix} \quad \text{gives} \quad \hat{\mathbf{x}} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}. \quad (8)$$

The combination $\mathbf{p} = A\hat{\mathbf{x}}$ is the projection of $\mathbf{b}$ onto the column space of A :

$$\mathbf{p} = 5 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - 3 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}. \quad \text{The error is} \quad \mathbf{e} = \mathbf{b} - \mathbf{p} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}. \quad (9)$$

Two checks on the calculation. First, the error $\mathbf{e} = (1, -2, 1)$ is perpendicular to both columns $(1, 1, 1)$ and $(0, 1, 2)$. Second, the final P times $\mathbf{b} = (6, 0, 0)$ correctly gives $\mathbf{p} = (5, 2, -1)$. That solves the problem for one particular $\mathbf{b}$.

To find $\mathbf{p} = P\mathbf{b}$ for every $\mathbf{b}$, compute $P = A(A^T A)^{-1} A^T$. The determinant of $A^T A$ is $15 - 9 = 6$; then $(A^T A)^{-1}$ is easy. Multiply A times $(A^T A)^{-1}$ times A^T to reach P :

$$(A^T A)^{-1} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} \quad \text{and} \quad P = \frac{1}{6} \begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}. \quad (10)$$

We must have $P^2 = P$, because a second projection doesn't change the first projection.

Warning The matrix $P = A(A^T A)^{-1} A^T$ is deceptive. You might try to split $(A^T A)^{-1}$ into A^{-1} times $(A^T)^{-1}$. If you make that mistake, and substitute it into P , you will find $P = AA^{-1}(A^T)^{-1} A^T$. Apparently everything cancels. This looks like $P = I$, the identity matrix. We want to say why this is wrong.

The matrix A is rectangular. It has no inverse matrix. We cannot split $(A^T A)^{-1}$ into A^{-1} times $(A^T)^{-1}$ because there is no A^{-1} in the first place.

In our experience, a problem that involves a rectangular matrix almost always leads to $A^T A$. When A has independent columns, $A^T A$ is invertible. This fact is so crucial that we state it clearly and give a proof.

$A^T A$ is invertible if and only if A has linearly independent columns.

Proof $A^T A$ is a square matrix (n by n). For every matrix A , we will now show that $A^T A$ has the same nullspace as A . When the columns of A are linearly independent, its nullspace contains only the zero vector. Then $A^T A$, with this same nullspace, is invertible.

Let A be any matrix. If $\mathbf{x}$ is in its nullspace, then $A\mathbf{x} = \mathbf{0}$. Multiplying by A^T gives $A^T A\mathbf{x} = \mathbf{0}$. So $\mathbf{x}$ is also in the nullspace of $A^T A$.

Now start with the nullspace of $A^T A$. From $A^T A\mathbf{x} = \mathbf{0}$ we must prove $A\mathbf{x} = \mathbf{0}$. We can't multiply by $(A^T)^{-1}$, which generally doesn't exist. Just multiply by $\mathbf{x}^T$:

$$(\mathbf{x}^T)A^T A\mathbf{x} = 0 \quad \text{or} \quad (A\mathbf{x})^T(A\mathbf{x}) = 0 \quad \text{or} \quad \|A\mathbf{x}\|^2 = 0.$$

This says: If $A^T A\mathbf{x} = \mathbf{0}$ then $A\mathbf{x}$ has length zero. Therefore $A\mathbf{x} = \mathbf{0}$. Every vector $\mathbf{x}$ in one nullspace is in the other nullspace. If $A^T A$ has dependent columns, so has A . If $A^T A$ has independent columns, so has A . This is the good case:

When A has independent columns, $A^T A$ is square, symmetric, and invertible.

To repeat for emphasis: $A^T A$ is $(n \text{ by } m)$ times $(m \text{ by } n)$. Then $A^T A$ is square $(n \text{ by } n)$. It is symmetric, because its transpose is $(A^T A)^T = A^T(A^T)^T$ which equals $A^T A$. We just proved that $A^T A$ is invertible—provided A has independent columns. Watch the difference between dependent and independent columns:

$$\begin{array}{ccc} A^T & A & A^T A \\ \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \end{bmatrix} & \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix} & = \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix} \\ \text{dependent} & \text{singular} & \end{array} \quad \begin{array}{ccc} A^T & A & A^T A \\ \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 1 \end{bmatrix} & = \begin{bmatrix} 2 & 4 \\ 4 & 9 \end{bmatrix} \\ \text{indep.} & \text{invertible} & \end{array}$$

Very brief summary To find the projection $\mathbf{p} = \hat{x}_1 \mathbf{a}_1 + \cdots + \hat{x}_n \mathbf{a}_n$, solve $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$. This gives $\hat{\mathbf{x}}$. The projection is $A\hat{\mathbf{x}}$ and the error is $\mathbf{e} = \mathbf{b} - \mathbf{p} = \mathbf{b} - A\hat{\mathbf{x}}$. The projection matrix $P = A(A^T A)^{-1}A^T$ gives $\mathbf{p} = P\mathbf{b}$.

This matrix satisfies $P^2 = P$. The distance from $\mathbf{b}$ to the subspace is $\|\mathbf{e}\|$.

■ REVIEW OF THE KEY IDEAS ■

1. The projection of $\mathbf{b}$ onto the line through $\mathbf{a}$ is $\mathbf{p} = a\hat{x} = a(a^T \mathbf{b} / a^T \mathbf{a})$.
2. The rank one projection matrix $P = \mathbf{a}\mathbf{a}^T / \mathbf{a}^T \mathbf{a}$ multiplies $\mathbf{b}$ to produce $\mathbf{p}$.
3. Projecting $\mathbf{b}$ onto a subspace leaves $\mathbf{e} = \mathbf{b} - \mathbf{p}$ perpendicular to the subspace.
4. When A has full rank n , the equation $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$ leads to $\hat{\mathbf{x}}$ and $\mathbf{p} = A\hat{\mathbf{x}}$.
5. The projection matrix $P = A(A^T A)^{-1}A^T$ has $P^T = P$ and $P^2 = P$.

■ WORKED EXAMPLES ■

4.2 A Project the vector $\mathbf{b} = (3, 4, 4)$ onto the line through $\mathbf{a} = (2, 2, 1)$ and then onto the plane that also contains $\mathbf{a}^* = (1, 0, 0)$. Check that the first error vector $\mathbf{b} - \mathbf{p}$ is perpendicular to $\mathbf{a}$, and the second error vector $\mathbf{e}^* = \mathbf{b} - \mathbf{p}^*$ is also perpendicular to $\mathbf{a}^*$.

Find the 3 by 3 projection matrix P onto that plane of $\mathbf{a}$ and $\mathbf{a}^*$. Find a vector whose projection onto the plane is the zero vector.

Solution The projection of $\mathbf{b} = (3, 4, 4)$ onto the line through $\mathbf{a} = (2, 2, 1)$ is $\mathbf{p} = 2\mathbf{a}$:

Onto a line
$$\mathbf{p} = \frac{\mathbf{a}^T \mathbf{b}}{\mathbf{a}^T \mathbf{a}} \mathbf{a} = \frac{18}{9} (2, 2, 1) = (4, 4, 2).$$

The error vector $\mathbf{e} = \mathbf{b} - \mathbf{p} = (-1, 0, 2)$ is perpendicular to $\mathbf{a}$. So $\mathbf{p}$ is correct.

The plane of $\mathbf{a} = (2, 2, 1)$ and $\mathbf{a}^* = (1, 0, 0)$ is the column space of $A = [\mathbf{a} \ \mathbf{a}^*]$:

$$A = \begin{bmatrix} 2 & 1 \\ 2 & 0 \\ 1 & 0 \end{bmatrix} \quad A^T A = \begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix} \quad (A^T A)^{-1} = \frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 9 \end{bmatrix} \quad P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & .8 & .4 \\ 0 & .4 & .2 \end{bmatrix}$$

Then $\mathbf{p}^* = P\mathbf{b} = (3, 4.8, 2.4)$. The error $\mathbf{e}^* = \mathbf{b} - \mathbf{p}^* = (0, -.8, 1.6)$ is perpendicular to $\mathbf{a}$ and $\mathbf{a}^*$. This $\mathbf{e}^*$ is in the nullspace of P and its projection is zero! Note $P^2 = P$.

4.2 B Suppose your pulse is measured at $x = 70$ beats per minute, then at $x = 80$, then at $x = 120$. Those three equations $Ax = \mathbf{b}$ in one unknown have $A^T = [1 \ 1 \ 1]$ and $\mathbf{b} = (70, 80, 120)$. *The best $\hat{x}$ is the _____ of 70, 80, 120.* Use calculus and projection:

1. Minimize $E = (x - 70)^2 + (x - 80)^2 + (x - 120)^2$ by solving $dE/dx = 0$.
2. Project $\mathbf{b} = (70, 80, 120)$ onto $\mathbf{a} = (1, 1, 1)$ to find $\hat{x} = \mathbf{a}^T \mathbf{b} / \mathbf{a}^T \mathbf{a}$.

Solution The closest horizontal line to the heights 70, 80, 120 is the *average* $\hat{x} = 90$:

$$\frac{dE}{dx} = 2(x - 70) + 2(x - 80) + 2(x - 120) = 0 \quad \text{gives} \quad \hat{x} = \frac{70 + 80 + 120}{3}$$

Projection :
$$\hat{x} = \frac{\mathbf{a}^T \mathbf{b}}{\mathbf{a}^T \mathbf{a}} = \frac{(1, 1, 1)^T (70, 80, 120)}{(1, 1, 1)^T (1, 1, 1)} = \frac{70 + 80 + 120}{3} = 90.$$

4.2 C In *recursive* least squares, a fourth measurement 130 changes $\hat{x}_{\text{old}}$ to $\hat{x}_{\text{new}}$. Compute $\hat{x}_{\text{new}}$ and verify the *update formula* $\hat{x}_{\text{new}} = \hat{x}_{\text{old}} + \frac{1}{4}(130 - \hat{x}_{\text{old}})$.

Going from 999 to 1000 measurements, $\hat{x}_{\text{new}} = \hat{x}_{\text{old}} + \frac{1}{1000}(b_{1000} - \hat{x}_{\text{old}})$ would only need $\hat{x}_{\text{old}}$ and the latest value b_{1000} . We don't have to average all 1000 numbers!

Solution The new measurement $b_4 = 130$ adds a fourth equation and $\hat{x}$ is updated to 100. You can average b_1, b_2, b_3, b_4 or combine the average of b_1, b_2, b_3 with b_4 :

$$\frac{70 + 80 + 120 + 130}{4} = 100 \quad \text{is also} \quad \hat{x}_{\text{old}} + \frac{1}{4}(b_4 - \hat{x}_{\text{old}}) = 90 + \frac{1}{4}(40).$$

The update from 999 to 1000 measurements shows the “gain matrix” $\frac{1}{1000}$ in a **Kalman filter** multiplying the prediction error $b_{\text{new}} - \hat{x}_{\text{old}}$. Notice $\frac{1}{1000} = \frac{1}{999} - \frac{1}{999000}$:

$$\hat{x}_{\text{new}} = \frac{b_1 + \cdots + b_{1000}}{1000} = \frac{b_1 + \cdots + b_{999}}{999} + \frac{1}{1000} \left(b_{1000} - \frac{b_1 + \cdots + b_{999}}{999} \right).$$

Problem Set 4.2

Questions 1–9 ask for projections onto lines. Also errors $e = b - p$ and matrices P .

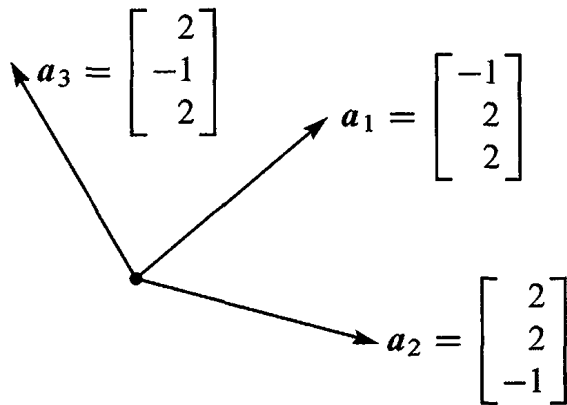
- 1 Project the vector b onto the line through a . Check that e is perpendicular to a :

$$(a) \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad \text{and} \quad a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad (b) \quad b = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \quad \text{and} \quad a = \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix}.$$

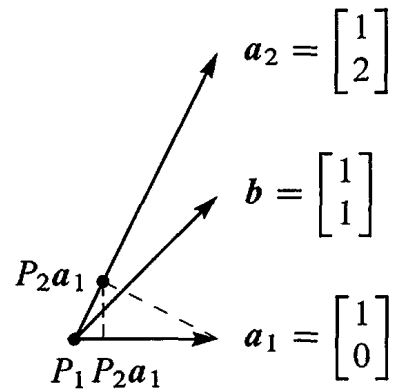
- 2 Draw the projection of b onto a and also compute it from $p = \hat{x}a$:

$$(a) \quad b = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \quad \text{and} \quad a = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (b) \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad a = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

- 3 In Problem 1, find the projection matrix $P = aa^T/a^T a$ onto the line through each vector a . Verify in both cases that $P^2 = P$. Multiply Pb in each case to compute the projection p .
- 4 Construct the projection matrices P_1 and P_2 onto the lines through the a 's in Problem 2. Is it true that $(P_1 + P_2)^2 = P_1 + P_2$? This *would* be true if $P_1 P_2 = 0$.
- 5 Compute the projection matrices $aa^T/a^T a$ onto the lines through $a_1 = (-1, 2, 2)$ and $a_2 = (2, 2, -1)$. Multiply those projection matrices and explain why their product $P_1 P_2$ is what it is.
- 6 Project $b = (1, 0, 0)$ onto the lines through a_1 and a_2 in Problem 5 and also onto $a_3 = (2, -1, 2)$. Add up the three projections $p_1 + p_2 + p_3$.
- 7 Continuing Problems 5–6, find the projection matrix P_3 onto $a_3 = (2, -1, 2)$. Verify that $P_1 + P_2 + P_3 = I$. The basis a_1, a_2, a_3 is orthogonal!
- 8 Project the vector $b = (1, 1)$ onto the lines through $a_1 = (1, 0)$ and $a_2 = (1, 2)$. Draw the projections p_1 and p_2 and add $p_1 + p_2$. The projections do not add to b because the a 's are not orthogonal.



Questions 5–6–7



Questions 8–9–10

- 9 In Problem 8, the projection of b onto the plane of a_1 and a_2 will equal b . Find $P = A(A^T A)^{-1} A^T$ for $A = [a_1 \ a_2] = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$.
- 10 Project $a_1 = (1, 0)$ onto $a_2 = (1, 2)$. Then project the result back onto a_1 . Draw these projections and multiply the projection matrices $P_1 P_2$: Is this a projection?

Questions 11–20 ask for projections, and projection matrices, onto subspaces.

- 11 Project b onto the column space of A by solving $A^T A \hat{x} = A^T b$ and $p = A \hat{x}$:

$$(a) \ A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \quad (b) \ A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}.$$

Find $e = b - p$. It should be perpendicular to the columns of A .

- 12 Compute the projection matrices P_1 and P_2 onto the column spaces in Problem 11. Verify that $P_1 b$ gives the first projection p_1 . Also verify $P_2^2 = P_2$.
- 13 (Quick and Recommended) Suppose A is the 4 by 4 identity matrix with its last column removed. A is 4 by 3. Project $b = (1, 2, 3, 4)$ onto the column space of A . What shape is the projection matrix P and what is P ?
- 14 Suppose b equals 2 times the first column of A . What is the projection of b onto the column space of A ? Is $P = I$ for sure in this case? Compute p and P when $b = (0, 2, 4)$ and the columns of A are $(0, 1, 2)$ and $(1, 2, 0)$.
- 15 If A is doubled, then $P = 2A(4A^T A)^{-1} 2A^T$. This is the same as $A(A^T A)^{-1} A^T$. The column space of $2A$ is the same as _____. Is $\hat{x}$ the same for A and $2A$?
- 16 What linear combination of $(1, 2, -1)$ and $(1, 0, 1)$ is closest to $b = (2, 1, 1)$?
- 17 (Important) If $P^2 = P$ show that $(I - P)^2 = I - P$. When P projects onto the column space of A , $I - P$ projects onto the _____.

- 18 (a) If P is the 2 by 2 projection matrix onto the line through $(1, 1)$, then $I - P$ is the projection matrix onto _____.
 (b) If P is the 3 by 3 projection matrix onto the line through $(1, 1, 1)$, then $I - P$ is the projection matrix onto _____.
- 19 To find the projection matrix onto the plane $x - y - 2z = 0$, choose two vectors in that plane and make them the columns of A . The plane should be the column space. Then compute $P = A(A^T A)^{-1} A^T$.
- 20 To find the projection matrix P onto the same plane $x - y - 2z = 0$, write down a vector e that is perpendicular to that plane. Compute the projection $Q = ee^T/e^T e$ and then $P = I - Q$.

Questions 21–26 show that projection matrices satisfy $P^2 = P$ and $P^T = P$.

- 21 Multiply the matrix $P = A(A^T A)^{-1} A^T$ by itself. Cancel to prove that $P^2 = P$. Explain why $P(Pb)$ always equals Pb : The vector Pb is in the column space so its projection is _____.
- 22 Prove that $P = A(A^T A)^{-1} A^T$ is symmetric by computing P^T . Remember that the inverse of a symmetric matrix is symmetric.
- 23 If A is square and invertible, the warning against splitting $(A^T A)^{-1}$ does not apply. It is true that $AA^{-1}(A^T)^{-1}A^T = I$. When A is invertible, why is $P = I$? What is the error e ?
- 24 The nullspace of A^T is _____ to the column space $C(A)$. So if $A^T b = 0$, the projection of b onto $C(A)$ should be $p = \underline{\hspace{2cm}}$. Check that $P = A(A^T A)^{-1} A^T$ gives this answer.
- 25 The projection matrix P onto an n -dimensional subspace has rank $r = n$. **Reason:** The projections Pb fill the subspace S . So S is the _____ of P .
- 26 If an m by m matrix has $A^2 = A$ and its rank is m , prove that $A = I$.
- 27 The important fact that ends the section is this: **If $A^T Ax = 0$ then $Ax = 0$.** **New Proof:** The vector Ax is in the nullspace of _____. Ax is always in the column space of _____. To be in both of those perpendicular spaces, Ax must be zero.
- 28 Use $P^T = P$ and $P^2 = P$ to prove that the length squared of column 2 always equals the diagonal entry P_{22} . This number is $\frac{2}{6} = \frac{4}{36} + \frac{4}{36} + \frac{4}{36}$ for

$$P = \frac{1}{6} \begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}.$$

- 29 If B has rank m (full row rank, independent rows) show that BB^T is invertible.

Challenge Problems

- 30** (a) Find the projection matrix P_C onto the column space of A (after looking closely at the matrix!)

$$A = \begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix}$$

- (b) Find the 3 by 3 projection matrix P_R onto the row space of A . Multiply $B = P_C A P_R$. Your answer B should be a little surprising—can you explain it?
- 31** In $\mathbf{R}^m$, suppose I give you $\mathbf{b}$ and $\mathbf{p}$, and $\mathbf{p}$ is a combination of $\mathbf{a}_1, \dots, \mathbf{a}_n$. How would you test to see if $\mathbf{p}$ is the projection of $\mathbf{b}$ onto the subspace spanned by the $\mathbf{a}$'s?
- 32** Suppose P_1 is the projection matrix onto the 1-dimensional subspace spanned by the first column of A . Suppose P_2 is the projection matrix onto the 2-dimensional column space of A . After thinking a little, compute the product $P_2 P_1$.

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

- 33** P_1 and P_2 are projections onto subspaces S and T . What is the requirement on those subspaces to have $P_1 P_2 = P_2 P_1$?

- 34** If A has r independent columns and B has r independent rows, AB is invertible.

Proof: When A is m by r with independent columns, we know that $A^T A$ is invertible. If B is r by n with independent rows, show that BB^T is invertible. (Take $A = B^T$.)

Now show that AB has rank r . Hint: Why does $A^T A B B^T$ have rank r ? That matrix multiplication by A^T and B^T cannot increase the rank of AB , by Problem 3.6:26.

4.3 Least Squares Approximations

It often happens that $Ax = b$ has no solution. The usual reason is: *too many equations*. The matrix has more rows than columns. There are more equations than unknowns (m is greater than n). The n columns span a small part of m -dimensional space. Unless all measurements are perfect, b is outside that column space. Elimination reaches an impossible equation and stops. But we can't stop just because measurements include noise.

To repeat: We cannot always get the error $e = b - Ax$ down to zero. When e is zero, x is an exact solution to $Ax = b$. When the length of e is as small as possible, $\hat{x}$ is a *least squares solution*. Our goal in this section is to compute $\hat{x}$ and use it. These are real problems and they need an answer.

The previous section emphasized p (the projection). This section emphasizes $\hat{x}$ (the least squares solution). They are connected by $p = A\hat{x}$. The fundamental equation is still $A^T A\hat{x} = A^T b$. Here is a short unofficial way to reach this equation:

When $Ax = b$ has no solution, multiply by A^T and solve $A^T A\hat{x} = A^T b$.

Example 1 A crucial application of least squares is fitting a straight line to m points. Start with three points: Find the closest line to the points $(0, 6)$, $(1, 0)$, and $(2, 0)$.

No straight line $b = C + Dt$ goes through those three points. We are asking for two numbers C and D that satisfy three equations. Here are the equations at $t = 0, 1, 2$ to match the given values $b = 6, 0, 0$:

$t = 0$	The first point is on the line $b = C + Dt$ if	$C + D \cdot 0 = 6$
$t = 1$	The second point is on the line $b = C + Dt$ if	$C + D \cdot 1 = 0$
$t = 2$	The third point is on the line $b = C + Dt$ if	$C + D \cdot 2 = 0$

This 3 by 2 system has *no solution*: $b = (6, 0, 0)$ is not a combination of the columns $(1, 1, 1)$ and $(0, 1, 2)$. Read off A , x , and b from those equations:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad x = \begin{bmatrix} C \\ D \end{bmatrix} \quad b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} \quad Ax = b \text{ is not solvable.}$$

The same numbers were in Example 3 in the last section. We computed $\hat{x} = (5, -3)$. Those numbers are the best C and D , so $5 - 3t$ will be the best line for the 3 points. We must connect projections to least squares, by explaining why $A^T A\hat{x} = A^T b$.

In practical problems, there could easily be $m = 100$ points instead of $m = 3$. They don't exactly match any straight line $C + Dt$. Our numbers 6, 0, 0 exaggerate the error so you can see e_1 , e_2 , and e_3 in Figure 4.6.

Minimizing the Error

How do we make the error $e = b - Ax$ as small as possible? This is an important question with a beautiful answer. The best x (called $\hat{x}$) can be found by geometry or algebra or calculus: 90° angle or project using P or set the derivative of the error to zero.

By geometry Every Ax lies in the plane of the columns $(1, 1, 1)$ and $(0, 1, 2)$. In that plane, we look for the point closest to b . *The nearest point is the projection p .*

The best choice for $A\hat{x}$ is p . The smallest possible error is $e = b - p$. The three points at heights (p_1, p_2, p_3) do lie on a line, because p is in the column space. In fitting a straight line, $\hat{x}$ gives the best choice for (C, D) .

By algebra Every vector b splits into two parts. The part in the column space is p . The perpendicular part in the nullspace of A^T is e . There is an equation we cannot solve ($Ax = b$). There is an equation $A\hat{x} = p$ we do solve (by removing e):

$$Ax = b = p + e \text{ is impossible; } A\hat{x} = p \text{ is solvable.} \quad (1)$$

The solution to $A\hat{x} = p$ leaves the least possible error (which is e):

$$\text{Squared length for any } x \quad \|Ax - b\|^2 = \|Ax - p\|^2 + \|e\|^2. \quad (2)$$

This is the law $c^2 = a^2 + b^2$ for a right triangle. The vector $Ax - p$ in the column space is perpendicular to e in the left nullspace. We reduce $Ax - p$ to zero by choosing x to be $\hat{x}$. That leaves the smallest possible error $e = (e_1, e_2, e_3)$.

Notice what “smallest” means. The *squared length* of $Ax - b$ is minimized:

The least squares solution $\hat{x}$ makes $E = \|Ax - b\|^2$ as small as possible.

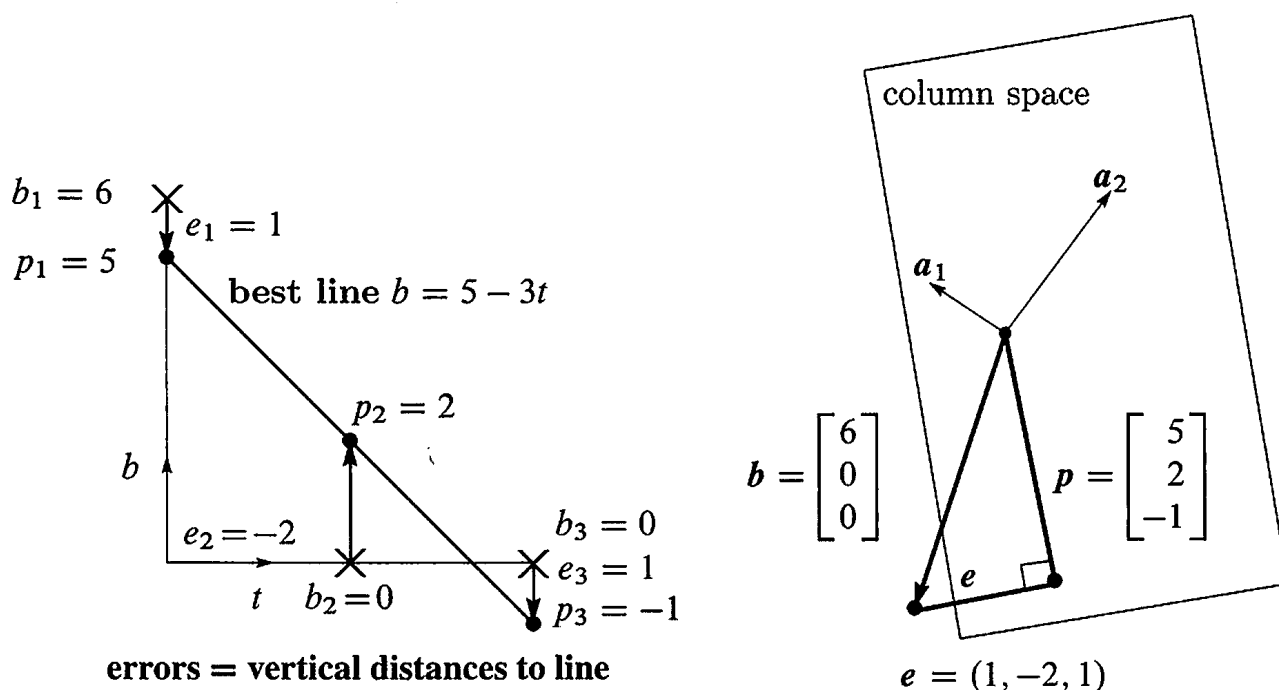


Figure 4.6: **Best line and projection: Two pictures, same problem.** The line has heights $p = (5, 2, -1)$ with errors $e = (1, -2, 1)$. The equations $A^T A\hat{x} = A^T b$ give $\hat{x} = (5, -3)$. The best line is $b = 5 - 3t$ and the projection is $p = 5a_1 - 3a_2$.

Figure 4.6a shows the closest line. It misses by distances $e_1, e_2, e_3 = 1, -2, 1$. Those are vertical distances. The least squares line minimizes $E = e_1^2 + e_2^2 + e_3^2$.

Figure 4.6b shows the same problem in 3-dimensional space ($\mathbf{b}$ p e space). The vector $\mathbf{b}$ is not in the column space of A . That is why we could not solve $A\mathbf{x} = \mathbf{b}$. No line goes through the three points. The smallest possible error is the perpendicular vector $\mathbf{e}$. This is $\mathbf{e} = \mathbf{b} - A\hat{\mathbf{x}}$, the vector of errors $(1, -2, 1)$ in the three equations. Those are the distances from the best line. Behind both figures is the fundamental equation $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$.

Notice that the errors $1, -2, 1$ add to zero. The error $\mathbf{e} = (e_1, e_2, e_3)$ is perpendicular to the first column $(1, 1, 1)$ in A . The dot product gives $e_1 + e_2 + e_3 = 0$.

By calculus Most functions are minimized by calculus! The graph bottoms out and the derivative in every direction is zero. Here the error function E to be minimized is a *sum of squares* $e_1^2 + e_2^2 + e_3^2$ (the square of the error in each equation):

$$E = \|A\mathbf{x} - \mathbf{b}\|^2 = (C + D \cdot 0 - 6)^2 + (C + D \cdot 1)^2 + (C + D \cdot 2)^2. \quad (3)$$

The unknowns are C and D . With two unknowns there are *two derivatives*—both zero at the minimum. They are “partial derivatives” because $\partial E / \partial C$ treats D as constant and $\partial E / \partial D$ treats C as constant:

$$\partial E / \partial C = 2(C + D \cdot 0 - 6) + 2(C + D \cdot 1) + 2(C + D \cdot 2) = 0$$

$$\partial E / \partial D = 2(C + D \cdot 0 - 6)(0) + 2(C + D \cdot 1)(1) + 2(C + D \cdot 2)(2) = 0.$$

$\partial E / \partial D$ contains the extra factors $0, 1, 2$ from the chain rule. (The last derivative from $(C + 2D)^2$ was 2 times $C + 2D$ times that extra 2.) In the C derivative the corresponding factors are $1, 1, 1$, because C is always multiplied by 1. It is no accident that $1, 1, 1$ and $0, 1, 2$ are the columns of A .

Now cancel 2 from every term and collect all C 's and all D 's:

$$\begin{array}{ll} \text{The } C \text{ derivative is zero: } 3C + 3D = 6 & \text{This matrix } \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \text{ is } A^T A \\ \text{The } D \text{ derivative is zero: } 3C + 5D = 0 & \end{array} \quad (4)$$

These equations are identical with $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$. The best C and D are the components of $\hat{\mathbf{x}}$. The equations from calculus are the same as the “normal equations” from linear algebra. These are the key equations of least squares:

The partial derivatives of $\|A\mathbf{x} - \mathbf{b}\|^2$ are zero when $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$.

The solution is $C = 5$ and $D = -3$. Therefore $b = 5 - 3t$ is the best line—it comes closest to the three points. At $t = 0, 1, 2$ this line goes through $\mathbf{p} = 5, 2, -1$. It could not go through $\mathbf{b} = 6, 0, 0$. The errors are $1, -2, 1$. This is the vector $\mathbf{e}$!

The Big Picture

The key figure of this book shows the four subspaces and the true action of a matrix. The vector $\mathbf{x}$ on the left side of Figure 4.3 went to $\mathbf{b} = A\mathbf{x}$ on the right side. In that figure $\mathbf{x}$ was split into $\mathbf{x}_r + \mathbf{x}_n$. There were many solutions to $A\mathbf{x} = \mathbf{b}$.

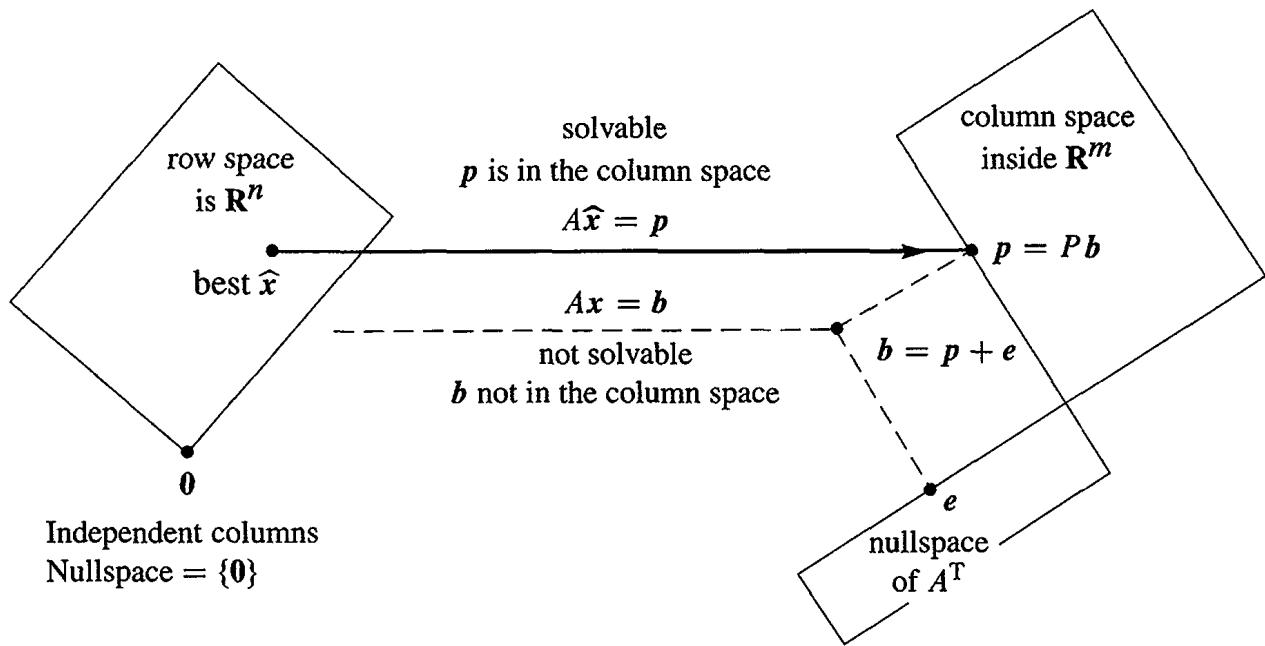


Figure 4.7: The projection $p = A\hat{x}$ is closest to b , so $\hat{x}$ minimizes $E = \|b - Ax\|^2$.

In this section the situation is just the opposite. There are *no* solutions to $Ax = b$. Instead of splitting up x we are splitting up b . Figure 4.3 shows the big picture for least squares. Instead of $Ax = b$ we solve $A\hat{x} = p$. The error $e = b - p$ is unavoidable.

Notice how the nullspace $N(A)$ is very small—just one point. With independent columns, the only solution to $Ax = 0$ is $x = 0$. Then $A^T A$ is invertible. The equation $A^T A\hat{x} = A^T b$ fully determines the best vector $\hat{x}$. The error has $A^T e = 0$.

Chapter 7 will have the complete picture—all four subspaces included. Every x splits into $x_r + x_n$, and every b splits into $p + e$. The best solution is $\hat{x}_r$ in the row space. We can't help e and we don't want x_n —this leaves $A\hat{x} = p$.

Fitting a Straight Line

Fitting a line is the clearest application of least squares. It starts with $m > 2$ points, hopefully near a straight line. At times $t_1, \dots, t_m$ those m points are at heights $b_1, \dots, b_m$. The best line $C + Dt$ misses the points by vertical distances $e_1, \dots, e_m$. No line is perfect, and the least squares line minimizes $E = e_1^2 + \dots + e_m^2$.

The first example in this section had three points in Figure 4.6. Now we allow m points (and m can be large). The two components of $\hat{x}$ are still C and D .

A line goes through the m points when we exactly solve $Ax = b$. Generally we can't do it. Two unknowns C and D determine a line, so A has only $n = 2$ columns. To fit the m points, we are trying to solve m equations (and we only want two!):

$$Ax = b \quad \text{is} \quad \begin{array}{l} C + Dt_1 = b_1 \\ C + Dt_2 = b_2 \\ \vdots \\ C + Dt_m = b_m \end{array} \quad \text{with} \quad A = \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}. \quad (5)$$

The column space is so thin that almost certainly $\mathbf{b}$ is outside of it. When $\mathbf{b}$ happens to lie in the column space, the points happen to lie on a line. In that case $\mathbf{b} = \mathbf{p}$. Then $A\mathbf{x} = \mathbf{b}$ is solvable and the errors are $\mathbf{e} = (0, \dots, 0)$.

The closest line $C + Dt$ has heights $p_1, \dots, p_m$ with errors $e_1, \dots, e_m$.

Solve $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ for $\hat{\mathbf{x}} = (C, D)$. The errors are $e_i = b_i - C - Dt_i$.

Fitting points by a straight line is so important that we give the two equations $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$, once and for all. The two columns of A are independent (unless all times t_i are the same). So we turn to least squares and solve $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$.

Dot-product matrix $A^T A = \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix}.$ (6)

On the right side of the normal equation is the 2 by 1 vector $A^T \mathbf{b}$:

$$A^T \mathbf{b} = \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix}. \quad (7)$$

In a specific problem, these numbers are given. The best $\hat{\mathbf{x}} = (C, D)$ is in equation (9).

The line $C + Dt$ minimizes $e_1^2 + \cdots + e_m^2 = \|A\mathbf{x} - \mathbf{b}\|^2$ when $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$:

$$\begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix}. \quad (8)$$

The vertical errors at the m points on the line are the components of $\mathbf{e} = \mathbf{b} - \mathbf{p}$. This error vector (the *residual*) $\mathbf{b} - A\hat{\mathbf{x}}$ is perpendicular to the columns of A (geometry). The error is in the nullspace of A^T (linear algebra). The best $\hat{\mathbf{x}} = (C, D)$ minimizes the total error E , the sum of squares:

$$E(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2 = (C + Dt_1 - b_1)^2 + \cdots + (C + Dt_m - b_m)^2.$$

When calculus sets the derivatives $\partial E / \partial C$ and $\partial E / \partial D$ to zero, it produces $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$.

Other least squares problems have more than two unknowns. Fitting by the best parabola has $n = 3$ coefficients C, D, E (see below). In general we are fitting m data points by n parameters $x_1, \dots, x_n$. The matrix A has n columns and $n < m$. The derivatives of $\|A\mathbf{x} - \mathbf{b}\|^2$ give the n equations $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$. **The derivative of a square is linear.** This is why the method of least squares is so popular.

Example 2 A has *orthogonal columns* when the measurement times t_i add to zero.

Suppose $b = 1, 2, 4$ at times $t = -2, 0, 2$. Those times add to zero. The columns of A have *zero dot product*:

$$\begin{array}{l} C + D(-2) = 1 \\ C + D(0) = 2 \\ C + D(2) = 4 \end{array} \quad \text{or} \quad Ax = \begin{bmatrix} 1 & -2 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

Look at the zeros in $A^T A$:

$$A^T A \hat{x} = A^T b \quad \text{is} \quad \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}.$$

Main point: Now $A^T A$ is *diagonal*. We can solve separately for $C = \frac{7}{3}$ and $D = \frac{6}{8}$. The zeros in $A^T A$ are dot products of perpendicular columns in A . The diagonal matrix $A^T A$, with entries $m = 3$ and $t_1^2 + t_2^2 + t_3^2 = 8$, is virtually as good as the identity matrix.

Orthogonal columns are so helpful that it is worth moving the time origin to produce them. To do that, subtract away the average time $\hat{t} = (t_1 + \cdots + t_m)/m$. The shifted times $T_i = t_i - \hat{t}$ add to $\sum T_i = m\hat{t} - m\hat{t} = 0$. With the columns now orthogonal, $A^T A$ is diagonal. Its entries are m and $T_1^2 + \cdots + T_m^2$. The best C and D have direct formulas:

$$T \text{ is } t - \hat{t} \quad C = \frac{b_1 + \cdots + b_m}{m} \quad \text{and} \quad D = \frac{b_1 T_1 + \cdots + b_m T_m}{T_1^2 + \cdots + T_m^2}. \quad (9)$$

The best line is $C + DT$ or $C + D(t - \hat{t})$. The time shift that makes $A^T A$ diagonal is an example of the Gram-Schmidt process: *orthogonalize the columns in advance*.

Fitting by a Parabola

If we throw a ball, it would be crazy to fit the path by a straight line. A parabola $b = C + Dt + Et^2$ allows the ball to go up and come down again (b is the height at time t). The actual path is not a perfect parabola, but the whole theory of projectiles starts with that approximation.

When Galileo dropped a stone from the Leaning Tower of Pisa, it accelerated. The distance contains a quadratic term $\frac{1}{2}gt^2$. (Galileo's point was that the stone's mass is not involved.) Without that t^2 term we could never send a satellite into the right orbit. But even with a nonlinear function like t^2 , the unknowns C, D, E appear linearly! Choosing the best parabola is still a problem in linear algebra.

Problem Fit heights $b_1, \dots, b_m$ at times $t_1, \dots, t_m$ by a parabola $C + Dt + Et^2$.

Solution With $m > 3$ points, the m equations for an exact fit are generally unsolvable:

$$\begin{array}{l} C + Dt_1 + Et_1^2 = b_1 \\ \vdots \\ C + Dt_m + Et_m^2 = b_m \end{array} \quad \text{has the } m \text{ by } 3 \text{ matrix} \quad A = \begin{bmatrix} 1 & t_1 & t_1^2 \\ \vdots & \vdots & \vdots \\ 1 & t_m & t_m^2 \end{bmatrix}. \quad (10)$$

Least squares The closest parabola $C + Dt + Et^2$ chooses $\hat{x} = (C, D, E)$ to satisfy the three normal equations $A^T A \hat{x} = A^T b$.

May I ask you to convert this to a problem of projection? The column space of A has dimension _____. The projection of $\mathbf{b}$ is $\mathbf{p} = A\hat{\mathbf{x}}$, which combines the three columns using the coefficients C, D, E . The error at the first data point is $e_1 = b_1 - C - Dt_1 - Et_1^2$. The total squared error is $e_1^2 + \dots$. If you prefer to minimize by calculus, take the partial derivatives of E with respect to _____, _____, _____. These three derivatives will be zero when $\hat{\mathbf{x}} = (C, D, E)$ solves the 3 by 3 system of equations _____.

Section 8.5 has more least squares applications. The big one is Fourier series—approximating functions instead of vectors. The function to be minimized changes from a sum of squared errors $e_1^2 + \dots + e_m^2$ to an integral of the squared error.

Example 3 For a parabola $b = C + Dt + Et^2$ to go through the three heights $b = 6, 0, 0$ when $t = 0, 1, 2$, the equations are

$$\begin{aligned} C + D \cdot 0 + E \cdot 0^2 &= 6 \\ C + D \cdot 1 + E \cdot 1^2 &= 0 \\ C + D \cdot 2 + E \cdot 2^2 &= 0. \end{aligned} \quad (11)$$

This is $A\mathbf{x} = \mathbf{b}$. We can solve it exactly. Three data points give three equations and a square matrix. The solution is $\mathbf{x} = (C, D, E) = (6, -9, 3)$. The parabola through the three points in Figure 4.8a is $b = 6 - 9t + 3t^2$.

What does this mean for projection? The matrix has three columns, which span the whole space $\mathbf{R}^3$. The projection matrix is the identity. The projection of $\mathbf{b}$ is $\mathbf{b}$. The error is zero. We didn't need $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$, because we solved $A\mathbf{x} = \mathbf{b}$. Of course we could multiply by A^T , but there is no reason to do it.

Figure 4.8 also shows a fourth point b_4 at time t_4 . If that falls on the parabola, the new $A\mathbf{x} = \mathbf{b}$ (four equations) is still solvable. When the fourth point is not on the parabola, we turn to $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$. Will the least squares parabola stay the same, with all the error at the fourth point? Not likely!

The smallest error vector (e_1, e_2, e_3, e_4) is perpendicular to $(1, 1, 1, 1)$, the first column of A . Least squares balances out the four errors, and they add to zero.

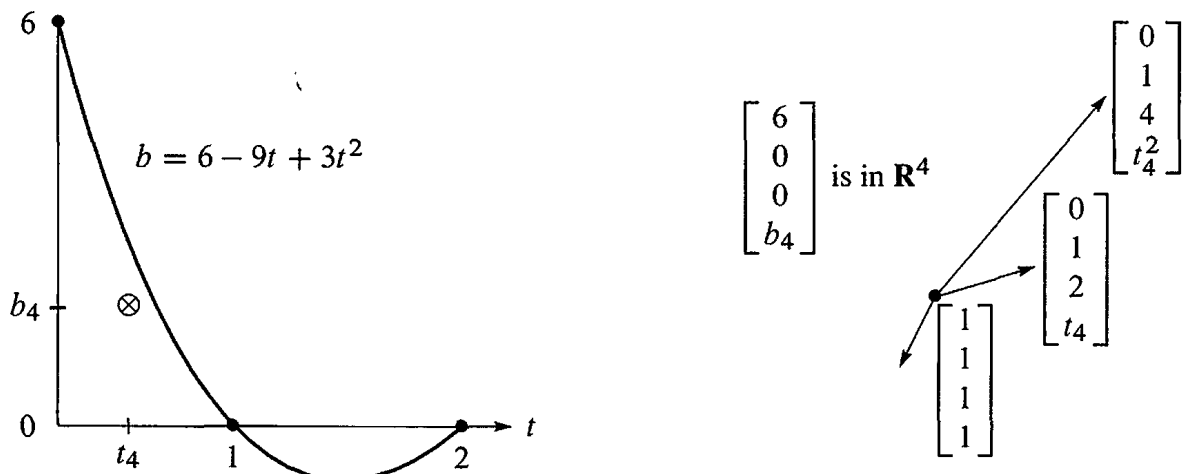


Figure 4.8: From Example 3: An exact fit of the parabola at $t = 0, 1, 2$ means that $\mathbf{p} = \mathbf{b}$ and $\mathbf{e} = \mathbf{0}$. The point b_4 off the parabola makes $m > n$ and we need least squares.

■ REVIEW OF THE KEY IDEAS ■

1. The least squares solution $\hat{\mathbf{x}}$ minimizes $E = \|\mathbf{Ax} - \mathbf{b}\|^2$. This is the sum of squares of the errors in the m equations ($m > n$).
2. The best $\hat{\mathbf{x}}$ comes from the normal equations $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$.
3. To fit m points by a line $b = C + Dt$, the normal equations give C and D .
4. The heights of the best line are $\mathbf{p} = (p_1, \dots, p_m)$. The vertical distances to the data points are the errors $\mathbf{e} = (e_1, \dots, e_m)$.
5. If we try to fit m points by a combination of $n < m$ functions, the m equations $\mathbf{Ax} = \mathbf{b}$ are generally unsolvable. The n equations $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ give the least squares solution—the combination with smallest MSE (mean square error).

■ WORKED EXAMPLES ■

4.3 A Start with nine measurements b_1 to b_9 , all zero, at times $t = 1, \dots, 9$. The tenth measurement $b_{10} = 40$ is an outlier. Find the best *horizontal line* $y = C$ to fit the ten points $(1, 0), (2, 0), \dots, (9, 0), (10, 40)$ using three measures for the error E :

- (1) Least squares $E_2 = e_1^2 + \dots + e_{10}^2$ (then the normal equation for C is linear)
- (2) Least maximum error $E_\infty = |e_{\max}|$
- (3) Least sum of errors $E_1 = |e_1| + \dots + |e_{10}|$.

Solution (1) The least squares fit to $0, 0, \dots, 0, 40$ by a horizontal line is $C = 4$:

$$A = \text{column of 1's} \quad A^T A = 10 \quad A^T \mathbf{b} = \text{sum of } b_i = 40. \quad \text{So } 10C = 40.$$

(2) The least maximum error requires $C = 20$, halfway between 0 and 40.

(3) The least sum requires $C = 0$ (!!). The sum of errors $9|C| + |40 - C|$ would increase if C moves up from zero.

The least sum comes from the *median* measurement (the median of $0, \dots, 0, 40$ is zero). Many statisticians feel that the least squares solution is too heavily influenced by outliers like $b_{10} = 40$, and they prefer least sum. But the equations become nonlinear.

Now find the least squares straight line $C + Dt$ through those ten points.

$$A^T A = \begin{bmatrix} 10 & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} = \begin{bmatrix} 10 & 55 \\ 55 & 385 \end{bmatrix} \quad A^T \mathbf{b} = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix} = \begin{bmatrix} 40 \\ 400 \end{bmatrix}$$

Those come from equation (8). Then $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ gives $C = -8$ and $D = 24/11$.

What happens to C and D if you multiply the b_i by 3 and then add 30 to get $\mathbf{b}_{\text{new}} = (30, 30, \dots, 150)$? Linearity allows us to rescale $\mathbf{b} = (0, 0, \dots, 40)$. Multiplying $\mathbf{b}$ by 3 will multiply C and D by 3. Adding 30 to all b_i will add 30 to C .

4.3 B Find the parabola $C + Dt + Et^2$ that comes closest (least squares error) to the values $\mathbf{b} = (0, 0, 1, 0, 0)$ at the times $t = -2, -1, 0, 1, 2$. First write down the five equations $A\mathbf{x} = \mathbf{b}$ in three unknowns $\mathbf{x} = (C, D, E)$ for a parabola to go through the five points. No solution because no such parabola exists. Solve $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$.

I would predict $D = 0$. Why should the best parabola be symmetric around $t = 0$? In $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$, equation 2 for D should uncouple from equations 1 and 3.

Solution The five equations $A\mathbf{x} = \mathbf{b}$ have a rectangular “Vandermonde” matrix A :

$$\begin{array}{rcl} C + D(-2) + E(-2)^2 & = & 0 \\ C + D(-1) + E(-1)^2 & = & 0 \\ C + D(0) + E(0)^2 & = & 1 \\ C + D(1) + E(1)^2 & = & 0 \\ C + D(2) + E(2)^2 & = & 0 \end{array} \quad A = \begin{bmatrix} 1 & -2 & 4 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \quad A^T A = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{bmatrix}$$

Those zeros in $A^T A$ mean that column 2 of A is orthogonal to columns 1 and 3. We see this directly in A (the times $-2, -1, 0, 1, 2$ are symmetric). The best C, D, E in the parabola $C + Dt + Et^2$ come from $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$, and D is uncoupled:

$$\begin{bmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{leads to} \quad \begin{array}{l} C = 34/70 \\ D = 0 \text{ as predicted} \\ E = -10/70 \end{array}$$

Problem Set 4.3

Problems 1–11 use four data points $\mathbf{b} = (0, 8, 8, 20)$ to bring out the key ideas.

- 1 With $\mathbf{b} = 0, 8, 8, 20$ at $t = 0, 1, 3, 4$, set up and solve the normal equations $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$. For the best straight line in Figure 4.9a, find its four heights p_i and four errors e_i . What is the minimum value $E = e_1^2 + e_2^2 + e_3^2 + e_4^2$?
- 2 (Line $C + Dt$ does go through p 's) With $\mathbf{b} = 0, 8, 8, 20$ at times $t = 0, 1, 3, 4$, write down the four equations $A\mathbf{x} = \mathbf{b}$ (unsolvable). Change the measurements to $p = 1, 5, 13, 17$ and find an exact solution to $A\hat{\mathbf{x}} = \mathbf{p}$.
- 3 Check that $\mathbf{e} = \mathbf{b} - \mathbf{p} = (-1, 3, -5, 3)$ is perpendicular to both columns of the same matrix A . What is the shortest distance $\|\mathbf{e}\|$ from $\mathbf{b}$ to the column space of A ?
- 4 (By calculus) Write down $E = \|A\mathbf{x} - \mathbf{b}\|^2$ as a sum of four squares—the last one is $(C + 4D - 20)^2$. Find the derivative equations $\partial E / \partial C = 0$ and $\partial E / \partial D = 0$. Divide by 2 to obtain the normal equations $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$.
- 5 Find the height C of the best *horizontal line* to fit $\mathbf{b} = (0, 8, 8, 20)$. An exact fit would solve the unsolvable equations $C = 0, C = 8, C = 8, C = 20$. Find the 4 by 1 matrix A in these equations and solve $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$. Draw the horizontal line at height $\hat{x} = C$ and the four errors in $\mathbf{e}$.

- 6 Project $\mathbf{b} = (0, 8, 8, 20)$ onto the line through $\mathbf{a} = (1, 1, 1, 1)$. Find $\hat{x} = \mathbf{a}^T \mathbf{b} / \mathbf{a}^T \mathbf{a}$ and the projection $\mathbf{p} = \hat{x} \mathbf{a}$. Check that $\mathbf{e} = \mathbf{b} - \mathbf{p}$ is perpendicular to $\mathbf{a}$, and find the shortest distance $\|\mathbf{e}\|$ from $\mathbf{b}$ to the line through $\mathbf{a}$.
- 7 Find the closest line $b = Dt$, *through the origin*, to the same four points. An exact fit would solve $D \cdot 0 = 0$, $D \cdot 1 = 8$, $D \cdot 3 = 8$, $D \cdot 4 = 20$. Find the 4 by 1 matrix and solve $A^T A \hat{x} = A^T \mathbf{b}$. Redraw Figure 4.9a showing the best line $b = Dt$ and the e 's.
- 8 Project $\mathbf{b} = (0, 8, 8, 20)$ onto the line through $\mathbf{a} = (0, 1, 3, 4)$. Find $\hat{x} = D$ and $\mathbf{p} = \hat{x} \mathbf{a}$. The best C in Problems 5–6 and the best D in Problems 7–8 do *not* agree with the best (C, D) in Problems 1–4. That is because $(1, 1, 1, 1)$ and $(0, 1, 3, 4)$ are _____ perpendicular.
- 9 For the closest parabola $b = C + Dt + Et^2$ to the same four points, write down the unsolvable equations $A\mathbf{x} = \mathbf{b}$ in three unknowns $\mathbf{x} = (C, D, E)$. Set up the three normal equations $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ (solution not required). In Figure 4.9a you are now fitting a parabola to 4 points—what is happening in Figure 4.9b?
- 10 For the closest cubic $b = C + Dt + Et^2 + Ft^3$ to the same four points, write down the four equations $A\mathbf{x} = \mathbf{b}$. Solve them by elimination. In Figure 4.9a this cubic now goes exactly through the points. What are $\mathbf{p}$ and $\mathbf{e}$?
- 11 The average of the four times is $\hat{t} = \frac{1}{4}(0 + 1 + 3 + 4) = 2$. The average of the four b 's is $\hat{b} = \frac{1}{4}(0 + 8 + 8 + 20) = 9$.
- (a) Verify that the best line goes through the center point $(\hat{t}, \hat{b}) = (2, 9)$.
- (b) Explain why $C + D\hat{t} = \hat{b}$ comes from the first equation in $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$.

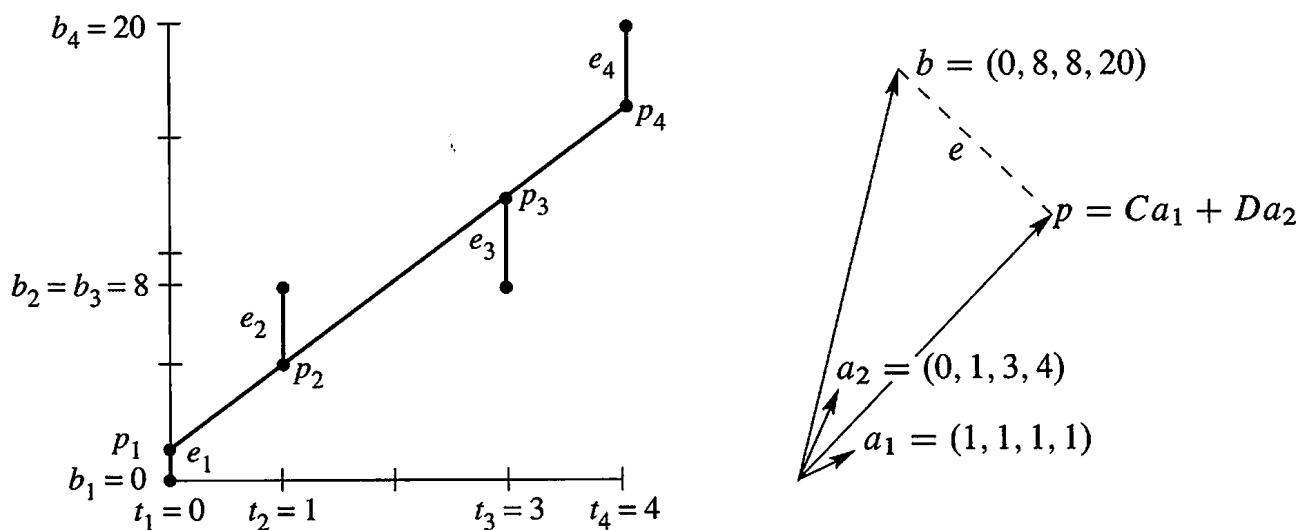


Figure 4.9: Problems 1–11: The closest line $C + Dt$ matches $Ca_1 + Da_2$ in $\mathbb{R}^4$.

Questions 12–16 introduce basic ideas of statistics—the foundation for least squares.

- 12 (Recommended) This problem projects $\mathbf{b} = (b_1, \dots, b_m)$ onto the line through $\mathbf{a} = (1, \dots, 1)$. We solve m equations $\mathbf{a}x = \mathbf{b}$ in 1 unknown (by least squares).
- (a) Solve $\mathbf{a}^T \mathbf{a} \hat{x} = \mathbf{a}^T \mathbf{b}$ to show that $\hat{x}$ is the *mean* (the average) of the $\mathbf{b}$'s.
 - (b) Find $\mathbf{e} = \mathbf{b} - \mathbf{a}\hat{x}$ and the *variance* $\|\mathbf{e}\|^2$ and the *standard deviation* $\|\mathbf{e}\|$.
 - (c) The horizontal line $\hat{b} = 3$ is closest to $\mathbf{b} = (1, 2, 6)$. Check that $\mathbf{p} = (3, 3, 3)$ is perpendicular to $\mathbf{e}$ and find the 3 by 3 projection matrix P .
- 13 First assumption behind least squares: $\mathbf{A}\mathbf{x} = \mathbf{b} - (\text{noise } \mathbf{e} \text{ with mean zero})$. Multiply the error vectors $\mathbf{e} = \mathbf{b} - \mathbf{A}\mathbf{x}$ by $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ to get $\hat{\mathbf{x}} - \mathbf{x}$ on the right. The estimation errors $\hat{\mathbf{x}} - \mathbf{x}$ also average to zero. The estimate $\hat{\mathbf{x}}$ is *unbiased*.
- 14 Second assumption behind least squares: The m errors e_i are independent with variance σ^2 , so the average of $(\mathbf{b} - \mathbf{A}\mathbf{x})(\mathbf{b} - \mathbf{A}\mathbf{x})^T$ is $\sigma^2 \mathbf{I}$. Multiply on the left by $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ and on the right by $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$ to show that the average matrix $(\hat{\mathbf{x}} - \mathbf{x})(\hat{\mathbf{x}} - \mathbf{x})^T$ is $\sigma^2 (\mathbf{A}^T \mathbf{A})^{-1}$. This is the *covariance matrix* P in section 8.6.
- 15 A doctor takes 4 readings of your heart rate. The best solution to $x = b_1, \dots, x = b_4$ is the average $\hat{x}$ of $b_1, \dots, b_4$. The matrix $\mathbf{A}$ is a column of 1's. Problem 14 gives the expected error $(\hat{x} - x)^2$ as $\sigma^2 (\mathbf{A}^T \mathbf{A})^{-1} = \underline{\hspace{2cm}}$. *By averaging, the variance drops from σ^2 to $\sigma^2/4$.*
- 16 If you know the average $\hat{x}_9$ of 9 numbers $b_1, \dots, b_9$, how can you quickly find the average $\hat{x}_{10}$ with one more number b_{10} ? The idea of *recursive* least squares is to avoid adding 10 numbers. What number multiplies $\hat{x}_9$ in computing $\hat{x}_{10}$?

$$\hat{x}_{10} = \frac{1}{10} b_{10} + \underline{\hspace{2cm}} \hat{x}_9 = \frac{1}{10} (b_1 + \dots + b_{10}) \quad \text{as in Worked Example 4.2 C.}$$

Questions 17–24 give more practice with $\hat{\mathbf{x}}$ and $\mathbf{p}$ and $\mathbf{e}$.

- 17 Write down three equations for the line $b = C + Dt$ to go through $b = 7$ at $t = -1$, $b = 7$ at $t = 1$, and $b = 21$ at $t = 2$. Find the least squares solution $\hat{\mathbf{x}} = (C, D)$ and draw the closest line.
- 18 Find the projection $\mathbf{p} = \mathbf{A}\hat{\mathbf{x}}$ in Problem 17. This gives the three heights of the closest line. Show that the error vector is $\mathbf{e} = (2, -6, 4)$. Why is $P\mathbf{e} = \mathbf{0}$?
- 19 Suppose the measurements at $t = -1, 1, 2$ are the errors 2, -6, 4 in Problem 18. Compute $\hat{\mathbf{x}}$ and the closest line to these new measurements. Explain the answer: $\mathbf{b} = (2, -6, 4)$ is perpendicular to $\underline{\hspace{2cm}}$ so the projection is $\mathbf{p} = \mathbf{0}$.
- 20 Suppose the measurements at $t = -1, 1, 2$ are $\mathbf{b} = (5, 13, 17)$. Compute $\hat{\mathbf{x}}$ and the closest line and $\mathbf{e}$. The error is $\mathbf{e} = \mathbf{0}$ because this $\mathbf{b}$ is $\underline{\hspace{2cm}}$.
- 21 Which of the four subspaces contains the error vector $\mathbf{e}$? Which contains $\mathbf{p}$? Which contains $\hat{\mathbf{x}}$? What is the nullspace of $\mathbf{A}$?

- 22** Find the best line $C + Dt$ to fit $b = 4, 2, -1, 0, 0$ at times $t = -2, -1, 0, 1, 2$.
- 23** Is the error vector e orthogonal to b or p or e or $\hat{x}$? Show that $\|e\|^2$ equals $e^T b$ which equals $b^T b - p^T b$. This is the smallest total error E .
- 24** The partial derivatives of $\|Ax\|^2$ with respect to $x_1, \dots, x_n$ fill the vector $2A^T Ax$. The derivatives of $2b^T Ax$ fill the vector $2A^T b$. So the derivatives of $\|Ax - b\|^2$ are zero when _____.

Challenge Problems

- 25** What condition on $(t_1, b_1), (t_2, b_2), (t_3, b_3)$ puts those three points onto a straight line? A column space answer is: (b_1, b_2, b_3) must be a combination of $(1, 1, 1)$ and (t_1, t_2, t_3) . Try to reach a specific equation connecting the t 's and b 's. I should have thought of this question sooner!
- 26** Find the plane that gives the best fit to the 4 values $b = (0, 1, 3, 4)$ at the corners $(1, 0)$ and $(0, 1)$ and $(-1, 0)$ and $(0, -1)$ of a square. The equations $C + Dx + Ey = b$ at those 4 points are $Ax = b$ with 3 unknowns $x = (C, D, E)$. What is A ? At the center $(0, 0)$ of the square, show that $C + Dx + Ey =$ average of the b 's.
- 27** (Distance between lines) The points $P = (x, x, x)$ and $Q = (y, 3y, -1)$ are on two lines in space that don't meet. Choose x and y to minimize the squared distance $\|P - Q\|^2$. The line connecting the closest P and Q is perpendicular to _____.
- 28** Suppose the columns of A are not independent. How could you find a matrix B so that $P = B(B^T B)^{-1} B^T$ does give the projection onto the column space of A ? (The usual formula will fail when $A^T A$ is not invertible.)
- 29** Usually there will be exactly one hyperplane in $\mathbf{R}^n$ that contains the n given points $x = 0, a_1, \dots, a_{n-1}$. (Example for $n = 3$: There will be one plane containing $0, a_1, a_2$ unless _____.) What is the test to have exactly one plane in $\mathbf{R}^n$?